Supplementary Material for CS-DRO

Seonggyeom Kim[*]
Dept. of Artificial Intelligence
Hanyang University
Seoul, South Korea
canino@hanyang.ac.kr

Eunjung Choi[*]
Dept. of Artificial Intelligence
Hanyang University
Seoul, South Korea
sinegi@hanyang.ac.kr

Dong-Kyu Chae
Dept. of Computer Science
Hanyang University
Seoul, South Korea
dongkyu@hanyang.ac.kr

Table 3: Summary of Notations

Symbol	Description
$\mathcal{X}, \mathcal{Y}$	Input space and label space ($\mathcal{Y} = \{1, \ldots, C\}$)
$\mathcal{Z}$	Latent representation space defined by ϕ
$\mathcal{V}$	Product space of feature and label ($\mathcal{V} := \mathcal{Z} \times \mathcal{Y}$)
$\mathcal{E}^{\text{src}}$	Set of source domains
(X^e, Y^e)	Random variables on $\mathcal{X} \times \mathcal{Y}$ for domain e
(x_i^e, y_i^e)	i-th sample from domain e
n^e	Number of samples in domain e
ϕ, w	Featurizer ($\mathcal{X} \to \mathcal{Z}$) and Classifier ($\mathcal{Z} \to \mathbb{R}^C$)
f_θ	Predictive model $f_\theta = w \circ \phi$
ℓ	Cross-entropy loss
B	A mini-batch of samples
n_B	Number of samples in a mini-batch
Z^e	Feature random variable induced by $\phi(X^e)$
Z_c	Domain-invariant (causal) feature component
Z_s^e	Domain-specific (spurious) feature component
P, Q	Probability distributions over $\mathcal{Z} \times \mathcal{Y}$
$\mathbf{U}^e$	Random vector in representation space $(Z^e, \tilde{Y}^e)$
G	Weighted adjacency matrix of the proxy DAG
G_Y	Mask vector derived from label-node edges in G
$\tilde{Y}$	Continuous variable for label in structural model
$c_G(v, v')$	Anisotropic transport cost aligned with G
$W_G(P, Q)$	Wasserstein distance induced by cost c_G
$\mathcal{L}_{\text{rec}}(\tilde{G}; Q)$	Reconstruction risk for graph $\tilde{G}$ under Q
$\mathcal{L}_G$	Structural regularization loss
$M_G(Q)$	Structural gradient signature of Q given G
$D_G(P, Q)$	Structure-preserving discrepancy between P, Q
ρ	Radius of the uncertainty set
κ	Structural budget for discrepancy D_G
$\mathcal{Q}_{W_G, D_G}(P)$	Causal structure-preserving uncertainty set
γ, β	Dual variables for constraints on ρ and κ

[*]Both authors contributed equally to this research.

A Notation

Table 3 summarizes the mathematical notations used in this paper.

B Structure-Preserving Discrepancy via Invariant Gradient Signatures

B.1 Definition and Computation of Causal Structural Signature and Structure-Preserving Discrepancy

Section 3.2 (main paper) used the gradient at $A = \mathbf{1}$ of the differentiable reconstruction risk $\mathcal{L}_{\text{rec}}(\tilde{G}; \cdot)$ (Eq. (6) of the main paper) as a structure-preserving signal, defining it as $M_G(\cdot)$ (Definition 3.3, Eq. (7) in the main paper). It also quantified the structural discrepancy between two distributions via the difference in M_G as D_G (Definition 3.4, Eq. (8) in the main paper). In this section, we expand M_G and D_G into forms convenient for subsequent analysis of their properties (in particular, linearity, continuity, and weak lower semi-continuity in Q). An important point in this expansion is that $M_G(Q)$ is a matrix-valued function of the same shape as A, and the expectation is taken entry-wise for each entry of this matrix.

As assumed in Section 3.2 (main paper), $\tilde{G}(\cdot)$ is linear in A with no additional nonlinearity. Therefore, $\tilde{G}(\cdot)$ can be treated as a linear operator in A, and the derivative of the squared error can be expanded via the standard chain rule. To simplify notation, for an arbitrary $v \in \mathbb{R}^{d+1}$ and matrix A, define the residual as $r(v, A) := v - \tilde{G}^\top v$. If we take the integrand (pointwise loss) in Eq. (6) (main paper) to be $\|r(v, A)\|_2^2$, the entry-wise partial derivative with respect to A is as follows:

$$\frac{\partial}{\partial A_{ji}} \|r(v, A)\|_2^2 = -2\, G_{ji}\, v_i\, r_j(v, A). \tag{27}$$

Eq. (27) specifically shows that $\nabla_A \|r(v, A)\|_2^2$ is a matrix of the same shape as A. That is, each element (j, i) consists of a combination term of the residual element $r_j(v, A)$ for a specific edge $(j \to i)$ and the input element v_j, selected edge-wise by G_{ji}. Collecting this into a matrix form, it is written as follows: $\nabla_A \|r(v, A)\|_2^2 = -2\Big(G \odot \big(v\, r(v, A)^\top\big)\Big)$, where $\nabla_A \|r(v, A)\|_2^2 \in \mathbb{R}^{(d+1)\times(d+1)}$, and $v\, r(v, A)^\top$ is a $(d+1) \times (d+1)$ rank-one matrix. Now, according to Definition 3.3 (Eq. (7)) of the main paper, $M_G(Q)$ for $Q \in \mathcal{P}(\mathcal{V})$ is expressed as follows: $M_G(Q) := \nabla_A \mathcal{L}_{\text{rec}}(\tilde{G}; Q)\big|_{A=\mathbf{1}} \in \mathbb{R}^{(d+1)\times(d+1)}$. In particular, each component of $M_G(Q)$ is defined as follows: $\big[M_G(Q)\big]_{ji} = \frac{\partial}{\partial A_{ji}} \mathbb{E}_{v\sim Q}\big[\|r(v, A)\|_2^2\big]\Big|_{A=\mathbf{1}}$. Under appropriate conditions to be specified in the subsequent lemma (e.g., differentiability and conditions justifying dominated convergence), the interchange of differentiation and expectation is allowed element-wise, so the following holds:

$$\left[M_G(Q)\right]_{ji} \;=\; \mathbb{E}_{v\sim Q}\left[\left.\frac{\partial}{\partial A_{ji}}\|r(v,A)\|_2^2\right|_{A=\mathbf{1}}\right]. \tag{28}$$

By stacking Eq. (28) for all (j,i), $M_G(Q)$ can be written as the *element-wise expectation* of the matrix-valued random variable $\nabla_A\|r(v,A)\|_2^2\big|_{A=\mathbf{1}}$ as follows:

$$M_G(Q) \;=\; \mathbb{E}_{v\sim Q}\left[\left.\nabla_A\|r(v,A)\|_2^2\right|_{A=\mathbf{1}}\right]. \tag{29}$$

Note that the result of the expectation in Eq. (29) is a matrix in $\mathbb{R}^{(d+1)\times(d+1)}$. That is, $\mathbb{E}_{v\sim Q}[\cdot]$ is applied to each element of the matrix. Eq. (29) provides a standard form for treating $M_G(Q)$ as a population-level functional with respect to Q. This form is particularly useful when discussing the linearity (or affine property), continuity, and weak lower semi-continuity properties of $M_G(Q)$. Also, as mentioned in Section 3.2 (main paper), for linear least-squares errors, $\mathcal{L}_{\mathrm{rec}}(\tilde{G};Q)$ is a convex function with respect to A, and the gradient matrix $M_G(Q)$ at $A=\mathbf{1}$ can be interpreted as a signal measuring the degree of violation of the first-order necessary condition under Q in an edge-wise manner. Finally, $D_G(P,Q)$ in Definition 3.4 (main paper) measures the difference of M_G using the Frobenius norm by Eq. (8) (main paper). Substituting Eq. (29), it can be expressed as follows:

$$\begin{aligned} D_G(P,Q) = \Big\| &\mathbb{E}_{v\sim P}\left[\left.\nabla_A\|r(v,A)\|_2^2\right|_{A=\mathbf{1}}\right] \\ &- \mathbb{E}_{v\sim Q}\left[\left.\nabla_A\|r(v,A)\|_2^2\right|_{A=\mathbf{1}}\right]\Big\|_F, \end{aligned} \tag{30}$$

Here, $\mathbb{E}_{v\sim Q}[\cdot]$ and $\mathbb{E}_{v\sim P}[\cdot]$ are matrix-valued respectively, and only $D_G(P,Q)$ ultimately results in a scalar through the Frobenius norm. Eq. (30) shows that it is a quantity directly comparing how the first-order necessary condition of the structure defined by G changes with distribution shift, using the difference of edge-wise gradient signatures. However, we do not interpret this comparison as comprehensively explaining arbitrary distribution shifts; in this paper, we utilize D_G only as a sufficient signal to serve as a structure-preserving constraint in the uncertainty set.

B.2 Convexity and Weak Lower Semi-Continuity of D_G

In this section, as required in Assumption 4.2(3) (main paper), we rigorously show that $D_G(\cdot,P)$ is convex with respect to $Q\in\mathcal{P}(\mathcal{V})$ and lower semi-continuous with respect to the weak topology of $\mathcal{P}(\mathcal{V})$. To this end, we reuse the development obtained in Supplementary Sec. B.1. The key consists of the following two steps. First, $M_G(Q)$ is by definition an A-gradient (matrix), and by the development in Supplementary Sec. B.1, it can be expressed as the element-wise expectation of the value at $A=\mathbf{1}$ of the matrix-valued random variable $\nabla_A\|r(v,A)\|_2^2$. Here, the interchange of differentiation and expectation must be justified for each element. Second, once the above representation is secured, $M_G(Q)$ becomes an affine functional with respect to Q, and under the compactness of $\mathcal{V}$, it becomes continuous with respect to weak convergence. Therefore, $D_G(P,Q)=\|M_G(P)-M_G(Q)\|_F$ becomes convex and weak lower semi-continuous.

Now, to treat $M_G(Q)$ as a population-level functional on $\mathcal{P}(\mathcal{V})$, we need to combine the definition in Definition 3.3 (main paper) with the element-wise differentiation development in Supplementary Sec. B.1 to justify the interchange of differentiation and expectation at $A=\mathbf{1}$. Since $M_G(Q)$ is a matrix, this interchange must be described element-wise for each element (j,i). The following Lemma is a standard justification step confirming this fact.

Lemma B.1 (Pointwise differentiation under expectation at $A=\mathbf{1}$). *Assume that under Assumption 4.1 (main paper), $\mathcal{V}\subset\mathbb{R}^{d+1}$ is compact, and as assumed in Section 3.2 (main paper), $\tilde{G}(\cdot)$ is linear in A and has no nonlinearity. For any $Q\in\mathcal{P}(\mathcal{V})$ and any (j,i), the following holds:*

$$\left.\frac{\partial}{\partial A_{ji}}\mathbb{E}_{v\sim Q}\left[\|r(v,A)\|_2^2\right]\right|_{A=\mathbf{1}} = \mathbb{E}_{v\sim Q}\left[\left.\frac{\partial}{\partial A_{ji}}\|r(v,A)\|_2^2\right|_{A=\mathbf{1}}\right]. \tag{31}$$

Therefore, $M_G(Q)$ can be expressed as follows:

$$M_G(Q) = \mathbb{E}_{v\sim Q}\left[\left.\nabla_A\|r(v,A)\|_2^2\right|_{A=\mathbf{1}}\right]. \tag{32}$$

Proof. Fix (j,i). According to the definition of partial differentiation, we use the difference quotient in the direction of edge E^{ji}. It appears as follows:

$$\begin{aligned} &\frac{\mathbb{E}_{v\sim Q}\left[\|r(v,\,\mathbf{1}+sE^{ji})\|_2^2\right]-\mathbb{E}_{v\sim Q}\left[\|r(v,\,\mathbf{1})\|_2^2\right]}{s} \\ &= \mathbb{E}_{v\sim Q}\left[\frac{\|r(v,\,\mathbf{1}+sE^{ji})\|_2^2-\|r(v,\,\mathbf{1})\|_2^2}{s}\right], \qquad s\neq 0. \end{aligned} \tag{33}$$

The left-hand side is the difference quotient defining the desired partial derivative, and the right-hand side is the form where the difference quotient is moved inside the expectation. Now, fix each $v\in\mathcal{V}$. Applying the mean value theorem in the s direction, there exists some $\theta=\theta(v,s)\in(0,1)$ such that:

$$\frac{\|r(v,\,1+sE^{ji})\|_2^2-\|r(v,\,1)\|_2^2}{s} = \left.\frac{\partial}{\partial A_{ji}}\|r(v,A)\|_2^2\right|_{A=\mathbf{1}+\theta sE^{ji}}. \tag{34}$$

Next, we establish a domination condition to interchange the integral and the limit. Fix $0<\varepsilon<1$ and consider only $|s|\le\varepsilon$. By the pointwise derivative (Eq. (27)) in Supplementary Sec. B.1, $\frac{\partial}{\partial A_{ji}}\|r(v,A)\|_2^2$ is continuous. Thus, from the compactness of $\mathcal{V}$ and the compactness in the ε-neighborhood of A, there exists a finite constant $C_{ji}<\infty$ such that the following holds:

$$\left|\left.\frac{\partial}{\partial A_{ji}}\|r(v,A)\|_2^2\right|_{A=\mathbf{1}+\theta sE^{ji}}\right| \le C_{ji}, \qquad \forall v\in\mathcal{V},\ \forall|s|\le\varepsilon. \tag{35}$$

Therefore, from Eq. (34), the difference quotient is dominated by the Q-integrable constant C_{ji}. Also, for each $v\in\mathcal{V}$, as $s\to 0$, by the definition of partial differentiation, it appears as follows:

$$\frac{\|r(v,\,1+sE^{ji})\|_2^2-\|r(v,\,1)\|_2^2}{s} \;\to\; \left.\frac{\partial}{\partial A_{ji}}\|r(v,A)\|_2^2\right|_{A=\mathbf{1}}. \tag{36}$$

Now, we apply the Lebesgue dominated convergence theorem to the right-hand side of Eq. (33). Eq. (35) provides the domination condition, and Eq. (36) provides the pointwise convergence. Thus, we can interchange the limit and expectation as $s\to 0$, obtaining Eq. (31). Finally, since Eq. (31) holds for arbitrary (j,i), collecting all elements establishes Eq. (32). $\square$

This is a standard justification step that connects the pointwise derivative calculated in Supplementary Sec. B.1 to match $M_G(Q)$ in the definition. Through this step, we can place $M_G(Q)$ in the form of an expectation with respect to distribution Q, and subsequently proceed rigorously with the argumentation for the convexity and weak lower semi-continuity of D_G at the definition level.

Through Lemma B.1, we have already obtained that $M_G(Q)$ is expressed as the expectation of the pointwise gradient matrix. Now, to show $M_G(Q_n) \to M_G(Q)$ under weak convergence, it suffices to show that the matrix given by the pointwise expansion in Supplementary Sec. B.1 is continuous and bounded on $\mathcal{V}$.

Lemma B.2 (Continuity and boundedness of the pointwise gradient matrix on $\mathcal{V}$)**.** *Assume that under Assumption 4.1 (main paper), $\mathcal{V}$ is compact, and that $\tilde{G}(\cdot)$ is linear with respect to A and has no additional nonlinearity. Then, the function $\nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}}$ with respect to v is continuous and bounded on V. That is, there exists a constant $C < \infty$ such that the following holds:*

$$\sup_{v\in\mathcal{V}} \Big\| \nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}} \Big\|_F \leq C. \tag{37}$$

Proof. Substituting the pointwise matrix expansion of Supplementary Sec. B.1 into $A = \mathbf{1}$, the pointwise gradient matrix appears as follows:

$$\nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}} = -2\Big(G \odot \big(r(v,\mathbf{1})\, v^\top\big)\Big). \tag{38}$$

First, we show continuity. Since $\tilde{G}(\cdot)$ is a finite-dimensional linear map, it is continuous, and thus $r(v,\mathbf{1}) = v - \tilde{G}(v)$ is continuous with respect to v. Also, $v\, r(v,1)^\top$ is continuous as it consists of element-wise multiplication and addition, and $G \odot (\cdot)$ is continuous as it is element-wise multiplication. Therefore, by Eq. (38), $\nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}}$ is continuous on V. Next, we show boundedness. Since the Frobenius norm is a continuous function given by polynomials and square roots of matrix elements, $\Big\| \nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}} \Big\|_F$ is continuous on V. Since $\mathcal{V}$ is compact, by the extreme value theorem, the above continuous function attains a maximum on V. Letting that maximum be C, Eq. (37) holds. ◻

Now, combining Lemma B.1 and Lemma B.2, we summarize the affine property and weak continuity of $M_G(Q)$. This result immediately leads to the convexity and weak lower semi-continuity of $D_G(P,Q) = \|M_G(Q) - M_G(P)\|_F$. First, we summarize the properties of M_G in a Lemma.

Lemma B.3 (Affine property and weak continuity of M_G)**.** *Under Assumption 4.1 (main paper), Lemma B.1, and Lemma B.2, the following holds: (i) for any $Q_1, Q_2 \in \mathcal{P}(\mathcal{V})$ and $\lambda \in [0,1]$, M_G is affine with respect to Q, (ii) if Q_n weakly converges to Q ($Q_n \Rightarrow Q$), then $M_G(Q_n) \to M_G(Q)$.*

Proof. (i) Applying the linearity of expectation to the conclusion of Lemma B.1 (Eq. (32)), the following holds: $M_G(\lambda Q_1 + (1-\lambda)Q_2) = \lambda M_G(Q_1) + (1-\lambda) M_G(Q_2)$. (ii) Assume $Q_n \Rightarrow Q$. By Lemma B.2, since $\nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}}$ is continuous and bounded on V, its component function $\big[\nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}}\big]_{ji}$ is also bounded continuous on V. Therefore, by the definition of weak convergence, the following holds for any (j,i):

$$\mathbb{E}_{v\sim Q_n}\Big[\big[\nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}}\big]_{ji}\Big] \to \mathbb{E}_{v\sim Q}\Big[\big[\nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1}}\big]_{ji}\Big]. \tag{39}$$

Combining Eq. (39) and the representation in Lemma B.1 (Eq. (32)), $M_G(Q_n)$ converges element-wise to $M_G(Q)$. Since the matrix space is finite-dimensional, pointwise convergence implies Frobenius norm convergence. Therefore, $M_G(Q_n) \to M_G(Q)$ holds. ◻

Lemma B.3 guarantees that $M_G(Q)$ responds linearly to distribution Q, and that $M_G(Q)$ does not change discontinuously when the distribution changes continuously in the weak topology. Now, using this property, we show the convexity and weak lower semi-continuity of D_G.

Lemma B.4 (Convexity and weak lower semi-continuity of $D_G(\cdot,P)$)**.** *Under Assumption 4.1 (main paper) and Lemma B.3, for any fixed $P \in \mathcal{P}(\mathcal{V})$, the following holds: (i) $D_G(\cdot,P)$ is convex on $\mathcal{P}(\mathcal{V})$, (ii) $D_G(\cdot,P)$ is lower semi-continuous with respect to the weak topology of $\mathcal{P}(\mathcal{V})$. Furthermore, $D_G(\cdot,P)$ is weakly continuous.*

Proof. (i) From Lemma B.3(i), the following holds:

$$\begin{aligned} &M_G(\lambda Q_1 + (1-\lambda)Q_2) - M_G(P) \\ &\qquad = \lambda\big(M_G(Q_1) - M_G(P)\big) + (1-\lambda)\big(M_G(Q_2) - M_G(P)\big). \end{aligned} \tag{40}$$

Applying the triangle inequality and positive homogeneity of the Frobenius norm to Eq. (40), the following holds:

$$D_G(\lambda Q_1 + (1-\lambda)Q_2, P) \leq \lambda D_G(Q_1,P) + (1-\lambda) D_G(Q_2,P). \tag{41}$$

Thus, convexity holds. (ii) Assume $Q_n \Rightarrow Q$. By Lemma B.3(ii), $M_G(Q_n) \to M_G(Q)$, and since the Frobenius norm is continuous, the following holds:

$$\begin{aligned} D_G(Q_n,P) = \|M_G(Q_n) - M_G(P)\|_F \to \|M_G(Q) - M_G(P)\|_F \\ = D_G(Q,P). \end{aligned} \tag{42}$$

Eq. (42) implies weak continuity, and since continuity implies lower semi-continuity, weak lower semi-continuity also holds. ◻

Lemma B.4 justifies that the convexity and weak lower semi-conti nuity of D_G required in Assumption 4.2(3) (main paper) actually follow from Assumption 4.1 (main paper) and the development in Supplementary Sec. B.1. In particular, since $D_G(\cdot,P)$ is continuous in the weak topology, the D_G-based constraint operates stably when taking the uncertainty set to be weakly closed.

Summarizing the above Lemmas, under Assumption 4.1 (main paper) (especially the compactness of $\mathcal{V}$) and the pointwise differentiation development of Supplementary Sec. B.1, $M_G(Q)$ is affine and weakly continuous with respect to Q. Therefore, $D_G(\cdot,P)$ is convex and lower semi-continuous with respect to the weak topology, and is in fact weakly continuous. This implies that the requirements of Assumption 4.2(3) (main paper) naturally hold in the setting of this paper.

B.3 Gradient Signatures

We summarize the perspective of interpreting Definition 3.3 and Definition 3.4 (Eq. (7)–(8)) in Section 3.2 of the main paper as structure-preserving signals. The key point is that the A-gradient at $A = \mathbf{1}$ directly reflects the first-order stationarity condition of the G-based reconstruction under a specific distribution, and comparing this signal across distributions leads directly to the structure-preserving discrepancy (Eq. (8) in the main paper).

This perspective connects with how DICD (Wang et al. [29]) handles environment-wise optimality. DICD points out that the constraint requiring the fit in each environment to be optimal involves an argmin structure and is thus difficult to handle directly, and to replace this with a differentiable surrogate, it inserts a multiplicative mask and then measures the first-order necessary condition (gradient = 0) at the all-ones matrix as a penalty. The theorem development in the linear/nonlinear settings of DICD clarifies that this surrogate is based on the logical structure that if the original optimality constraint holds, the first-order necessary condition at the all-ones matrix must follow.

Now, we map this perspective to the gated reconstruction setting of this paper. In Section 3.2 (main paper), we view the reconstruction risk of Eq. (6) (main paper) as a function of the gate A, and define the gradient matrix at $A = \mathbf{1}$ as Eq. (7) (main paper). Under the assumptions of Section 3.2 in the main paper (specifically linearity with respect to A and squared loss), Eq. (6) is convex and differentiable with respect to A, so if $A = \mathbf{1}$ is a local minimum at $\hat{P}$, the following holds by the standard first-order necessary condition: $\nabla_A \mathcal{L}_{\text{rec}}(\tilde{G}; \hat{P})\big|_{A=\mathbf{1}} = 0$. Therefore, Eq. (6) (main paper) summarizes at the matrix level whether the G-based reconstruction under $\hat{P}$ satisfies stationarity at $A = \mathbf{1}$. Also, the pointwise expansion in Supplementary Sec. B.1 (e.g., Eq. (38)) concretizes that this gradient can be interpreted edge-wise.

If the value of Eq. (7) (main paper) is non-zero, $A = \mathbf{1}$ is not a stationary point, which implies that there is room for the loss in Eq. (6) (main paper) to decrease from the perspective of first-order approximation by small changes around that point. To quantify this, the Frobenius norm of the gradient at $A = \mathbf{1}$ is interpreted as a value summarizing the maximum directional derivative: $\|M_G(Q)\|_F = \max_{\|\Delta\|_F=1} \langle M_G(Q), \Delta \rangle$. Therefore, the magnitude of Eq. (7) (main paper) indicates the maximum possibility of first-order reduction around $A = \mathbf{1}$, and in this paper, we interpret this as a first-order residual for how much the G-induced reconstruction violates the stationarity condition on distribution Q. This interpretation shares the same directionality as the way DICD penalizes the gradient at the all-ones matrix as the degree of violation of the optimality constraint.

Finally, Definition 3.4 (Eq. (8)) of the main paper directly compares the difference in the values of Eq. (7) between two distributions. That is, Eq. (8) quantifies how much the first-order optimality condition at $A = \mathbf{1}$ defined by G has changed due to distribution shift. In the next subsection, based on this perspective, we theoretically formalize that Eq. (8) operates as a structure-preserving constraint.

B.4 Theoretical Justification of Structure-Preserving Constraint

We summarize that the D_G constraint defined by Definition 3.4 (Eq. (8)) of the main paper can serve as a structure-preserving constraint. The key logic lies in the perspective of first-order optimality. Specifically, viewing the reconstruction risk in Eq. (6) as a function of the gate A, if the gradient residual (Eq. (7) at $A = \mathbf{1}$ is small, the local improvement obtainable by gate perturbation in the neighborhood of $A = \mathbf{1}$ is limited. Therefore, the condition that Eq. (8) is small can be interpreted as a constraint implying that the stationarity of G learned on P is maintained on distribution Q.

It should be emphasized here that the neighborhood radius α introduced below is a *scale for local analysis in A-space*, and has a different role from the budget κ used in the uncertainty set. κ is the magnitude of the constraint on the distribution pair (Q, P), whereas α is an auxiliary parameter to quantify the perturbation size in the neighborhood of $A = \mathbf{1}$.

Proposition B.5 (Local improvement bound induced by Definition 3.4)**.** *We follow the setting of Section 3.2 (main paper). In particular, we assume that $\mathcal{L}_{\text{rec}}$ in Eq. (6) is convex and differentiable with respect to the gate A. For any distribution $Q \in \mathcal{P}(\mathcal{V})$ and $\alpha > 0$, consider the following Frobenius neighborhood: $\mathcal{A}_\alpha := \{A : \|A - \mathbf{1}\|_F \le \alpha\}$. Then, the following inequality holds.*

$$\inf_{A \in \mathcal{A}_\alpha} \mathcal{L}_{\text{rec}}(\tilde{G}; Q) \;\ge\; \mathcal{L}_{\text{rec}}(\tilde{G}; Q)\big|_{A=\mathbf{1}} \;-\; \alpha \, \|M_G(Q)\|_F. \tag{43}$$

Also, from the triangle inequality and Eq. (8), the following holds.

$$\|M_G(Q)\|_F \;\le\; D_G(P, Q) + \|M_G(P)\|_F. \tag{44}$$

Thus, combining Eq. (43) *and Eq.* (44) *yields:*

$$\inf_{A \in \mathcal{A}_\alpha} \mathcal{L}_{\text{rec}}(\tilde{G}; Q) \;\ge\; \mathcal{L}_{\text{rec}}(\tilde{G}; Q)\big|_{A=\mathbf{1}} \;-\; \alpha\Big(D_G(P, Q) + \|M_G(P)\|_F\Big). \tag{45}$$

Proof. For notational simplicity, let $\varphi(A) := \mathcal{L}_{\text{rec}}(\tilde{G}; Q)\big|_A$. By the convexity and differentiability of φ, the following holds for all A:

$$\varphi(A) \;\ge\; \varphi(\mathbf{1}) + \langle \nabla_A \varphi(\mathbf{1}), A - \mathbf{1} \rangle \,. \tag{46}$$

By Eq. (7), $\nabla_A \varphi(\mathbf{1}) = M_G(Q)$. Applying the Cauchy–Schwarz inequality yields:

$$\langle M_G(Q), A - \mathbf{1} \rangle \;\ge\; -\|M_G(Q)\|_F \, \|A - \mathbf{1}\|_F. \tag{47}$$

Combining Eq. (46) and Eq. (47) and restricting to $A \in \mathcal{A}_\alpha$, we obtain Eq. (43). Next, from the triangle inequality, the following holds:

$$\|M_G(Q)\|_F \;\le\; \|M_G(Q) - M_G(P)\|_F + \|M_G(P)\|_F. \tag{48}$$

Since $\|M_G(Q) - M_G(P)\|_F = D_G(P, Q)$ by Eq. (8), Eq. (48) is equivalent to Eq. (44). Finally, substituting Eq. (44) into Eq. (43) yields Eq. (45). □

Corollary B.6 (Interpretation as a structure-preserving constraint)**.** *If G is well-trained on P such that $\|M_G(P)\|_F \le \varepsilon_P$ holds, and also $D_G(P, Q) \le \kappa$, then the following holds.*

$$\|M_G(Q)\|_F \;\le\; \kappa + \varepsilon_P. \tag{49}$$

Therefore, for all perturbations $A \in \mathcal{A}_\alpha$ in the neighborhood of $A = \mathbf{1}$, the value of Eq. (6) can decrease by at most $\alpha(\kappa + \varepsilon_P)$ compared to the value at $A = \mathbf{1}$. In this sense, $D_G(P, Q) \le \kappa$ can be interpreted as a structure-preserving constraint that maintains the first-order stationarity of G learned on P even on Q.

Proof. Substituting $\|M_G(P)\|_F \le \varepsilon_P$ and $D_G(P, Q) \le \kappa$ into Eq. (44) yields Eq. (49). The subsequent conclusion holds by substituting the same upper bound into Eq. (45). □

Theorem B.7 (Vanishing discrepancy under invariant local optimality at $A = \mathbf{1}$)**.** *Assume that there exists an oracle graph $G^\star$ such that for all distributions $\nu \in \mathcal{P}_{G^\star}$ belonging to the structure-preserving family $\mathcal{P}_{G^\star}$, $A = \mathbf{1}$ is a local minimizer of the A-problem in Eq. (6). Then, for all $\nu \in \mathcal{P}_{G^\star}$, the following holds.*

$$M_{G^\star}(\nu) = 0. \tag{50}$$

Therefore, for any $P, Q \in \mathcal{P}_{G^\star}$, the following holds.

$$D_{G^\star}(P,Q) = 0. \tag{51}$$

Proof. Arbitrarily fix $\nu \in \mathcal{P}_{G^\star}$. Since $A = \mathbf{1}$ is a local minimizer and the function is differentiable, by the standard first-order necessary condition, the following holds:

$$\nabla_A \mathcal{L}_{\mathrm{rec}}(\tilde{G}^\star; \nu)\big|_{A=1} = 0. \tag{52}$$

By Eq. (7), Eq. (52) implies $M_{G^\star}(\nu) = 0$. Thus, Eq. (50) holds. Now, applying Eq. (8) yields:

$$D_{G^\star}(P,Q) = \|M_{G^\star}(Q) - M_{G^\star}(P)\|_F = \|0-0\|_F = 0, \tag{53}$$

This is equivalent to Eq. (51). □

In practice, the optimal G is unknown, so an estimate $\hat{G}$ is used. In this case, how accurately $D_{\hat{G}}$ approximates D_G becomes important, which reduces to how sensitive M_G is with respect to G. Below, we directly bound the graph-stability of M_G using the pointwise representation (e.g., Eq. (38)) in Supplementary Sec. B.1, and consequently obtain a stability inequality for D_G.

Proposition B.8 (Stability bound under graph estimation)**.** *Assume that under Assumption 4.1 (main paper), $\mathcal{V}$ is compact, and also that the set of candidate graphs is compact such that there exists some $\mathrm{B}_G < \infty$ where $\|G\|_F \le \mathrm{B}_G$ for all admissible G. Also, let $\mathrm{B}_V := \sup_{v\in\mathcal{V}} \|v\|_2$. Then, for any $\nu \in \mathcal{P}(\mathcal{V})$ and any $G, \hat{G}$, the following holds:*

$$\|M_{\hat{G}}(\nu) - M_G(\nu)\|_F \;\le\; 2(1+2\mathrm{B}_G)\mathrm{B}_V^2\,\|\hat{G} - G\|_F. \tag{54}$$

Therefore, for any $P, Q \in \mathcal{P}(\mathcal{V})$, the following holds:

$$\left|D_{\hat{G}}(P,Q) - D_G(P,Q)\right| \;\le\; 4(1+2\mathrm{B}_G)\mathrm{B}_V^2\,\|\hat{G} - G\|_F. \tag{55}$$

Proof. Using the pointwise expansion (Eq. (38)) in Supplementary Sec. B.1, $M_G(\nu)$ is expressed as the expectation of a matrix-valued function of v, and this matrix consists of element-wise products involving G and $r(v,1)$. Also, as assumed in Section 3.2 (main paper), since the reconstruction map is a finite-dimensional linear operator induced by $\tilde{G}$, the residual at $A = \mathbf{1}$ varies linearly with respect to G. Using this to decompose $M_{\hat{G}}(\nu) - M_G(\nu)$ into a term for $(\hat{G} - G)$ and a term for the residual difference, and then sequentially applying the Frobenius bound for element-wise products and the Cauchy–Schwarz inequality, we obtain:

$$\|M_{\hat{G}}(\nu) - M_G(\nu)\|_F \;\le\; 2\|\hat{G} - G\|_F\,(1+2\mathrm{B}_G)\,\mathbb{E}_{v\sim\nu}\big[\|v\|_2^2\big]. \tag{56}$$

Finally, since $\mathbb{E}_{v\sim\nu}[\|v\|_2^2] \le \mathrm{B}_V^2$ holds from the compactness of $\mathcal{V}$, we obtain Eq. (54). Now, from the 1-Lipschitz property of the norm and the triangle inequality, the following holds:

$$\left|D_{\hat{G}}(P,Q) - D_G(P,Q)\right| \;\le\; \|M_{\hat{G}}(Q) - M_G(Q)\|_F + \|M_{\hat{G}}(P) - M_G(P)\|_F. \tag{57}$$

Applying Eq. (54) for $\nu = Q$ and $\nu = P$ yields Eq. (55). □

Remark. Eq. (55) shows that the "graph-stability constant of D_G" used in Lemma E.1 of the main text can be chosen concretely. That is, L_M in Lemma E.1 can be set as follows: $L_M := 4(1+2\mathrm{B}_G)\mathrm{B}_V^2$. Therefore, when using $\hat{G}$, even if $D_{\hat{G}}(P,Q) \le \kappa$ is enforced, the true discrepancy $D_G(P,Q)$ may be loosened by an error proportional to $\|\hat{G} - G\|_F$. From this perspective, Eq. (55) quantitatively suggests how the budget κ should be corrected when constructing the uncertainty set based on $\hat{G}$.

C Proofs of Theoretical Results

C.1 Proof of Proposition 4.4

Proof of Proposition 4.4 (main paper). In this proof, we show that the uncertainty set defined in Definition 3.5 (main paper) is nonempty, convex, and weakly compact. We also show that the inner worst-case risk in Eq. (11) is actually attained on $\mathcal{Q}_{W_G,D_G}(P)$. First, when the uncertainty set is given as in Definition 3.5 (main paper), $\mathcal{P}(\mathcal{V})$ is the set of Borel probability measures on V. Also, the weak topology is induced by standard weak convergence $Q_n \Rightarrow Q$, which implies that $\int \varphi\, dQ_n \to \int \varphi\, dQ$ holds for any bounded continuous function φ.

Step 1. If $\rho \ge 0$ and $\kappa \ge 0$, it suffices to show that $Q = P$ is immediately feasible. Since $c_G(v,v) = 0$ in Assumption 4.2(2), the transport cost for the identical distribution pair becomes 0, and thus $W_G(P,P) = 0$. Also, since D_G is defined such that the structural discrepancy is 0 for the identical distribution pair, $D_G(P,P) = 0$. Consequently, $P \in \mathcal{Q}_{W_G,D_G}(P)$, and $\mathcal{Q}_{W_G,D_G}(P) \ne \emptyset$.

Step 2. Let us take arbitrary $Q_1, Q_2 \in \mathcal{Q}_{W_G,D_G}(P)$ and $\lambda \in [0,1]$, and let $Q_\lambda := \lambda Q_1 + (1-\lambda)Q_2$. To show that $\mathcal{Q}_{W_G,D_G}(P)$ is convex, it suffices to show that $\{Q \in \mathcal{P}(\mathcal{V}) : D_G(P,Q) \le \kappa\}$ and $\{Q \in \mathcal{P}(\mathcal{V}) : W_G(P,Q) \le \rho\}$ are each convex. First, by Assumption 4.2(3), $D_G(\cdot, P)$ is convex with respect to Q, so the following holds:

$$D_G(Q_\lambda, P) \le \lambda D_G(Q_1, P) + (1-\lambda) D_G(Q_2, P). \tag{58}$$

Next, we show the convexity of $\{Q : W_G(P,Q) \le \rho\}$. Since W_G is the type-2 Wasserstein distance, it can be expressed as follows:

$$W_G(P,Q)^2 = \inf_{\pi\in\Pi(Q,P)} \mathbb{E}_{(v,v')\sim\pi}\big[c_G(v,v')\big]. \tag{59}$$

If we take couplings π_i from Q_i to P respectively, then $\pi_\lambda := \lambda\pi_1 + (1-\lambda)\pi_2$ becomes a valid coupling having Q_λ and P as marginals. Also, since the expectation of the transport cost is linear with respect to the coupling, taking the infimum in the definition yields the following inequality:

$$W_G(Q_\lambda, P)^2 \le \lambda W_G(Q_1,P)^2 + (1-\lambda) W_G(Q_2,P)^2. \tag{60}$$

Now, since Q_1, Q_2 satisfy $W_G(Q_i,P) \le \rho$ and $D_G(Q_i,P) \le \kappa$ respectively, $D_G(Q_\lambda,P) \le \kappa$ holds from Eq. (58). Also, if $W_G(Q_i,P) \le \rho$, then $W_G(Q_i,P)^2 \le \rho^2$, and since $W_G(Q_\lambda,P)^2 \le \rho^2$ from Eq. (60), $W_G(Q_\lambda,P) \le \rho$ holds. Therefore, $Q_\lambda \in \mathcal{Q}_{W_G,D_G}(P)$, and $\mathcal{Q}_{W_G,D_G}(P)$ is convex with respect to the constraints.

Step 3. The key points are (i) the weak compactness of $\mathcal{P}(\mathcal{V})$ and (ii) the weak closedness of $\mathcal{Q}_{W_G,D_G}(P)$. First, by Assumption 4.1, since $\mathcal{Z}$ is compact and $\mathcal{Y}$ is finite, $\mathcal{V} = \mathcal{Z} \times \mathcal{Y}$ is a compact metric space. In this case, every probability measure on V is tight, and according to Prokhorov's theorem [3, 22], $\mathcal{P}(\mathcal{V})$ is compact in the weak topology. Now, let us show that $\mathcal{Q}_{W_G,D_G}(P)$ is weakly closed in $\mathcal{P}(\mathcal{V})$. By Assumption 4.2(2)–(3), when viewing $W_G(P,Q)$ and $D_G(P,Q)$ as functions of Q, they are respectively lower semi-continuous with respect to weak convergence. Generally, the sublevel sets of a lower semi-continuous function are closed. Therefore, the following two sets are weakly closed:

$$\{Q \in \mathcal{P}(\mathcal{V}) : W_G(P,Q) \le \rho\}\ , \qquad \{Q \in \mathcal{P}(\mathcal{V}) : D_G(P,Q) \le \kappa\}\,.$$

Thus, their intersection, $\mathcal{Q}_{W_G,D_G}(P)$, is also weakly closed. In conclusion, since it is a weakly closed subset of the weakly compact space $\mathcal{P}(\mathcal{V})$, $\mathcal{Q}_{W_G,D_G}(P)$ is weakly compact.

Step 4. For any fixed $\theta \in \Theta$, we define the expected loss under distribution Q as follows:

$$\mathcal{L}_\theta(Q) := \mathbb{E}_{(z,y)\sim Q}\big[\ell(w(z),y)\big]. \tag{61}$$

By Assumption 4.2(1), $\ell(w(z),y)$ is upper semi-continuous and lower bounded on V. Also, since $\mathcal{V}$ is compact, an upper semi-continuous function attains a maximum on V, and in particular, is bounded above. Therefore, the definition in Eq. (61) has a well-defined finite value for all $Q \in \mathcal{P}(\mathcal{V})$. Now, when $Q_n \Rightarrow Q$, the following holds by the Portmanteau theorem for bounded upper semi-continuous functions:

$$\limsup_{n\to\infty} \int \ell(w(z),y)\,dQ_n \;\le\; \int \ell(w(z),y)\,dQ. \tag{62}$$

Therefore, $\mathcal{L}_\theta(\cdot)$ is upper semi-continuous in the weak topology. Now, let the worst-case value be:

$$\overline{\mathcal{L}}_\theta := \sup_{Q\in\mathcal{Q}_{W_G,D_G}(P)} \mathcal{L}_\theta(Q). \tag{63}$$

By the definition of Eq. (63), we can choose a maximizing sequence $\{Q_n\}_{n\ge1} \subset \mathcal{Q}_{W_G,D_G}(P)$ such that $\mathcal{L}_\theta(Q_n) \uparrow \overline{\mathcal{L}}_\theta$. Since $\mathcal{Q}_{W_G,D_G}(P)$ is weakly compact from Step 3, there exists a subsequence (denoted as Q_n again for convenience) such that $Q_n \Rightarrow Q^\star$ and satisfies $Q^\star \in \mathcal{Q}_{W_G,D_G}(P)$. At this time, from Eq. (62), we obtain the following:

$$\overline{\mathcal{L}}_\theta = \lim_{n\to\infty} \mathcal{L}_\theta(Q_n) \;\le\; \mathcal{L}_\theta(Q^\star) \;\le\; \sup_{Q\in\mathcal{Q}_{W_G,D_G}(P)} \mathcal{L}_\theta(Q) = \overline{\mathcal{L}}_\theta. \tag{64}$$

Therefore, $\mathcal{L}_\theta(Q^\star) = \overline{\mathcal{L}}_\theta$, and the supremum is attained at $Q^\star$. That is, there exists some $Q^\star(\theta) \in \mathcal{Q}_{W_G,D_G}(P)$ that actually attains the inner supremum of Eq. (11).

In summary, $\mathcal{Q}_{W_G,D_G}(P)$ is nonempty, convex, and weakly compact, and the inner worst-case risk is attained on $\mathcal{Q}_{W_G,D_G}(P)$. ◻

C.2 Proof of Proposition 4.5

Proof of Proposition 4.5 (main paper). Proposition 4.5 shows that the inner constrained supremum in Eq. (11) satisfies strong duality, and the Lagrangian relaxation to Eq. (12)–(13) can coincide at the distribution level. That is, the dual representation of Eq. (21) holds, implying a zero duality gap. In the following, we proceed with the discussion for arbitrarily fixed theta and proxy DAG G.

Step 1. The uncertainty set is as in Definition 3.5 (main paper). Since $W_G(P,Q) = W_2(Q,P) \ge 0$, the following equivalence holds:

$$W_G(P,Q) \le \rho \quad\Longleftrightarrow\quad W_G(P,Q)^2 \le \rho^2. \tag{65}$$

Therefore, the inner worst-case risk can be equivalently written as: $\sup_{Q\in P(V):\, W_G(P,Q)^2\le\rho^2,\, D_G(P,Q)\le\kappa} \mathcal{L}_\theta(Q)$. Also, we set the expected loss under distribution Q as follows, following the notation in Supplementary Sec. C.1 (Eq. (61)). $\mathcal{L}_\theta(Q) := E_{(z,y)\sim Q}\Big[\ell\big(w(z),y\big)\Big]$. Now, we define the Lagrangian function used in Eq. (12) (for the equivalent constraint Eq. (65)) as follows.

$$R(Q,\gamma,\beta;\theta) := \mathcal{L}_\theta(Q) - \gamma\Big(W_G(P,Q)^2 - \rho^2\Big) - \beta\Big(D_G(P,Q) - \kappa\Big). \tag{66}$$

Here, γ and β are restricted to be non-negative. Consider the following value for a fixed Q: $\inf_{\gamma\ge0,\,\beta\ge0} R(Q,\gamma,\beta;\theta)$. Here, if Q belongs to the uncertainty set, both constraint violations are less than or equal to 0, so taking $\gamma = 0$ and $\beta = 0$ makes the value of the above inf exactly $\mathcal{L}_\theta(Q)$. Conversely, if Q is outside the uncertainty set, some constraint has a positive violation, so sending the corresponding dual variable to infinity causes $R(Q,\gamma,\beta;\theta)$ to diverge to negative infinity. Thus, for infeasible Q, the above inf value becomes negative infinity. From this observation, infeasible Q is automatically excluded from the outer sup, so the following equality holds.

$$\sup_{Q\in\mathcal{Q}_{W_G,D_G}(P)} \mathcal{L}_\theta(Q) = \sup_{Q\in P(V)} \inf_{\gamma\ge0,\,\beta\ge0} R(Q,\gamma,\beta;\theta). \tag{67}$$

Step 2. To obtain Eq. (21), we need to interchange sup and inf in the above equation. To this end, we apply Sion's minimax theorem [25]. First, by Assumption 4.1, $\mathcal{V}$ is a compact metric space, and according to Prokhorov's theorem, $\mathcal{P}(\mathcal{V})$ is weakly compact in the weak topology. Also, since $\mathcal{P}(\mathcal{V})$ is closed under convex combinations of probability measures, it is a compact convex set in the weak topology. Next, we check the functional properties with respect to Q for fixed (γ,β). As shown in Step 4 of Supplementary Sec. B.1, by Assumption 4.2(1), $\ell(w(z),y)$ is upper semi-continuous and lower bounded on V. Also, since $\mathcal{V}$ is compact, $\ell(w(z),y)$ attains a maximum on V, and in particular, is bounded above. Therefore, $\ell(w(z),y)$ is a bounded upper semi-continuous function, and by the Portmanteau theorem (Eq. (62) in Supplementary Sec. C.1), $\mathcal{L}_\theta(Q)$ is upper semi-continuous in the weak topology. Also, $\mathcal{L}_\theta(Q)$ is a linear function with respect to Q. By Assumption 4.2(2)–(3), since $W_G(P,Q)^2$ and $D_G(P,Q)$ are convex in Q and lower semi-continuous in the weak topology, for $(\gamma,\beta) \succeq 0$, $R(Q,\gamma,\beta;\theta)$ is concave in Q and upper semi-continuous in the weak topology. Conversely, if we fix Q, $R(Q,\gamma,\beta;\theta)$ is affine (hence convex) and continuous with respect to (γ,β). Also, the domain of the dual variables $\{(\gamma,\beta) : \gamma \ge 0,\ \beta \ge 0\}$ is convex. Finally, since strict feasibility holds by Assumption 4.3 (main paper) (e.g., $Q = P$ is strictly feasible), we can apply Sion's minimax theorem to exchange $\sup_Q \inf_{\gamma,\beta} = \inf_{\gamma,\beta} \sup_Q$. Therefore, the following holds.

$$\sup_{Q\in P(V)} \inf_{\gamma\ge0,\,\beta\ge0} R(Q,\gamma,\beta;\theta) = \inf_{\gamma\ge0,\,\beta\ge0} \sup_{Q\in P(V)} R(Q,\gamma,\beta;\theta). \tag{68}$$

Step 3. Expanding the definition of $R(\cdot)$, the following holds for fixed (γ,β).

$$\begin{aligned}\sup_{Q\in P(V)} R(Q,\gamma,\beta;\theta) &= \gamma\rho^2 + \beta\kappa \\ &\quad + \sup_{Q\in P(V)} \Big(\mathcal{L}_\theta(Q) - \gamma W_G(P,Q)^2 - \beta D_G(P,Q)\Big).\end{aligned} \tag{69}$$

Combining Step 1 and Step 2, we finally obtain the following.

$$\begin{aligned}&\sup_{Q\in\mathcal{Q}_{W_G,D_G}(P)} \mathcal{L}_\theta(Q) \\ &\qquad = \inf_{\gamma\ge0,\,\beta\ge0} \Big\{\gamma\rho^2 + \beta\kappa \\ &\qquad\quad + \sup_{Q\in P(V)} \big(\mathcal{L}_\theta(Q) - \gamma W_G(P,Q)^2 - \beta D_G(P,Q)\big)\Big\}.\end{aligned} \tag{70}$$

This corresponds to the dual representation of Proposition 4.5 under the equivalent transformation Eq. (65). Thus, the inner problem of Eq. (11) satisfies strong duality and has a zero duality gap. Consequently, Eq. (13) can be justified as a primal–dual objective for distribution-level robustness. □

C.3 Proof of Theorem 4.7

Proof. The proof proceeds by (i) reducing the inclusion to a deviation event between P and $\hat{P}_n$ using the triangle inequality, and then combining (ii) the geometric concentration for $W_G(P,\hat{P}_n)$ and (iii) the structural concentration for $D_G(P,\hat{P}_n)$.

Step 1. The event to be shown in Theorem 4.7 (main paper) is $P^T \in \mathcal{Q}_{W_G,D_G}(\hat{P}_n)$, where $\mathcal{Q}_{W_G,D_G}(\cdot)$ is given by Eq. (10). This is equivalent to $W_G(P^T,\hat{P}_n) \le \rho$ and $D_G(P^T,\hat{P}_n) \le \kappa$ holding simultaneously. Also, by Assumption 4.6, $W_G(P,P^T) \le \rho_0$ and $D_G(P,P^T) \le \kappa_0$ hold. First, applying the triangle inequality to $W_G(\cdot,\cdot)$, we can express it as follows: $W_G(P^T,\hat{P}_n) \le W_G(P^T,P) + W_G(P,\hat{P}_n) \le \rho_0 + W_G(P,\hat{P}_n)$. Next, for D_G, since $D_G(P,Q) = \|M_G(Q) - M_G(P)\|_F$ in Eq. (8), and $M_G(\cdot)$ is a matrix-valued functional defined by Eq. (7), we can express it as follows using the triangle inequality of the Frobenius norm:

$$\begin{aligned} D_G(P^T,\hat{P}_n) =& \|M_G(P^T) - M_G(\hat{P}_n)\|_F \le \\ & \|M_G(P^T) - M_G(P)\|_F + \|M_G(P) - M_G(\hat{P}_n)\|_F \le \\ & \kappa_0 + D_G(P,\hat{P}_n). \end{aligned} \tag{71}$$

Therefore, (when $\rho > \rho_0$) if the following two conditions hold simultaneously, target inclusion follows:

$$W_G(P,\hat{P}_n) \le \rho - \rho_0, \qquad D_G(P,\hat{P}_n) \le \kappa - \kappa_0. \tag{72}$$

Now, we define the following for subsequent developments:

$$\Delta_\rho = \frac{\rho - \rho_0}{2}, \qquad \Delta_\kappa = \frac{\kappa - \kappa_0}{2} \tag{73}$$

Then, it suffices for Eq. (72) that $W_G(P,\hat{P}_n) \le 2\Delta_\rho$ and $D_G(P,\hat{P}_n) \le 2\Delta_\kappa$. Thus, from the union bound, we can express it as follows:

$$\begin{aligned} \mathbb{P}\Big(P^T \in \mathcal{Q}_{W_G,D_G}(\hat{P}_n)\Big) \ge 1 - \mathbb{P}\Big(W_G(P,\hat{P}_n) > 2\Delta_\rho\Big) \\ - \mathbb{P}\Big(D_G(P,\hat{P}_n) > 2\Delta_\kappa\Big). \end{aligned} \tag{74}$$

We now upper bound each of the two deviation terms.

Step 2. Since c_G in Eq. (9) (main paper) assigns an ∞ penalty to label mismatch, the effective geometry of W_G is dominated by the G_Y-aligned direction. Also, since $\mathcal{Z}$ is compact by Assumption 4.1, for convenience, let us define the following G_Y-aligned diameter:

$$\mathrm{diam}_G := \sup_{z,z' \in \mathcal{Z}} \|G_Y \odot (z - z')\|_2 < \infty. \tag{75}$$

We also define the (dimensionless scale) covering number for the $(\mathcal{Z}, \| G_Y \odot (\cdot) \|_2)$-geometry as follows:

$$\begin{aligned} N_G(\delta) := \min\Big\{& m \ge 1 : \exists \xi_1,\dots,\xi_m \in \mathcal{Z} \\ & \text{s.t. } \mathcal{Z} \subseteq \cup_{j=1}^m \{z : \|G_Y \odot (z - \xi_j)\|_2 \le \delta\,\mathrm{diam}_G\}\Big\}, \\ & \delta \in (0,1]. \end{aligned} \tag{76}$$

Now, applying standard discretization and union bound arguments for empirical measure concentration [7], we obtain the following upper bound when choosing the covering scale as $\delta = \Delta_\rho/(2\,\mathrm{diam}_G)$:

$$\mathbb{P}\Big(W_G(P,\hat{P}_n) > 2\Delta_\rho\Big) \le 2^{N_G(\Delta_\rho/(2\mathrm{diam}_G))+1} \cdot \exp\left(-n \cdot \frac{2\Delta_\rho^2}{\mathrm{diam}_G^2}\right). \tag{77}$$

Therefore, in the geometric term of Theorem 4.7, we can set:

$$\mathrm{C}(\Delta_\rho) := 2^{N_G(\Delta_\rho/(2\mathrm{diam}_G))+1}, \qquad \mathrm{a}(\Delta_\rho) := \frac{2\Delta_\rho^2}{\mathrm{diam}_G^2}. \tag{78}$$

Here, $\mathrm{C}(\Delta_\rho)$ is a covering-number-type complexity induced by the c_G-geometry (i.e., the G_Y-aligned subspace), and $\mathrm{a}(\Delta_\rho)$ is an exponential rate determined by the boundedness and diameter parameter.

Step 3. By Eq. (8) and the development in Supplementary Sec. B.1–B.2 (specifically Lemma B.1 and Lemma B.2), the pointwise gradient matrix is uniformly bounded on $\mathcal{V} = \mathcal{Z} \times \mathcal{Y}$. To this end, we set the following finite constant:

$$\mathrm{H}_{\max} := \sup_{v \in \mathcal{V}} \Big\| \nabla_A \|r(v,A)\|_2^2\big|_{A=1} \Big\|_F < \infty. \tag{79}$$

Now, since $V_1,\dots,V_n \sim P$ i.i.d. and $\hat{P}_n = \frac{1}{n}\sum_{i=1}^n \delta_{V_i}$, applying the exchange result of Lemma B.1, $M_G(\hat{P}_n)$ is expressed as the average of i.i.d. matrix-valued random variables. Therefore, for $D_G(P,\hat{P}_n)$, the change when replacing only one sample is dominated by $\mathrm{H}_{\max}$. That is, since the bounded-differences condition holds, we can apply McDiarmid's inequality [17], and for all $t > 0$, it can be expressed as follows:

$$\mathbb{P}\Big(D_G(P,\hat{P}_n) > t\Big) \le \exp\left(-n\,\frac{t^2}{8\mathrm{H}_{\max}^2}\right). \tag{80}$$

Thus, substituting $t = 2\Delta_\kappa$, we obtain the following:

$$\mathbb{P}\Big(D_G(P,\hat{P}_n) > 2\Delta_\kappa\Big) \le \exp\left(-n \cdot \frac{\Delta_\kappa^2}{2\mathrm{H}_{\max}^2}\right). \tag{81}$$

That is, in the structural term of Theorem 4.7, we can set:

$$\mathrm{b}(\Delta_\kappa) := \frac{\Delta_\kappa^2}{2\mathrm{H}_{\max}^2}. \tag{82}$$

Step 4. Finally, combining Eq. (77) and Eq. (81) into Eq. (74), it can be expressed as follows:

$$\mathbb{P}\Big(P^T \in \mathcal{Q}_{W_G,D_G}(\hat{P}_n)\Big) \ge 1 - \mathrm{C}(\Delta_\rho)e^{-n\mathrm{a}(\Delta_\rho)} - e^{-n\mathrm{b}(\Delta_\kappa)}. \tag{83}$$

Also, if $\rho > \rho_0$ and $\kappa > \kappa_0$, then $\Delta_\rho > 0$ and $\Delta_\kappa > 0$, so in Eq. (78) and Eq. (82), $\mathrm{a}(\Delta_\rho) > 0$ and $\mathrm{b}(\Delta_\kappa) > 0$, and thus the right-hand side converges exponentially to 1 with respect to n. □

D Approximation Gaps between Theory and Implementation

The theoretical analysis in this paper targets the CS-DRO objective having an inner supremum at the distribution-level. On the other hand, actual training stochastically approximates and solves the inner maximization via J-step gradient ascent on mini-batches to realistically train deep neural networks. We define the error that may exist between the theoretical objective function and the objective function for stochastic optimization.

D.1 Error Definitions

The theory of this paper (Section 4 of the main paper) includes the inner problem of the CS-DRO objective defined on the empirical distribution $\hat{P}_n$ or distribution P in the form of sup. However, the implementation of Algorithm 1 (main paper) replaces the distribution-level inner sup problem with a mini-batch B-based optimization problem for the computational feasibility of deep neural network training, and solves it by approximating with J-step gradient ascent (refer to Eq. (16) in the main text).

In this section, we define the approximation error occurring between the "ideal inner problem defined at the distribution level" and the "mini-batch based, J-step approximate inner problem used in implementation" in terms of two terms. The first term is the error when the batch-level inner problem optimization is not sufficiently performed due to the J-step approximation, and the second term is the error representing the phenomenon where the objective function value fluctuates due to the randomness of mini-batch sampling.

Batch-level inner objective. Let a mini-batch $B = \{(z_i, y_i)\}_{i=1}^{n_B}$ be given. Also, consider a mini-batch $B' = \{(z_i', y_i)\}_{i=1}^{n_B}$ where only the representations are perturbed while maintaining the labels of each sample. Similar to Eq. (16) in the main text, we set the inner objective function value defined as a function of B' when the mini-batch B and parameters (θ, γ, β) are fixed, as follows:

$$\begin{aligned}\mathcal{L}_{\text{in}}(B'; B, \theta, \gamma, \beta) :=& \frac{1}{n_B} \sum_{i=1}^{n_B} \ell\big(w_\theta(G_Y \odot z_i'), y_i\big) \\ &- \frac{\gamma}{n_B} \sum_{i=1}^{n_B} \|G_Y \odot (z_i - z_i')\|_2^2 - \beta\, D_G(B', B).\end{aligned} \tag{84}$$

Here, θ is the model parameter, and (γ, β) are dual variables corresponding to each constraint. Also, G_Y and $D_G(\cdot,\cdot)$ follow the definitions in the main text (Section 3.4.3).

D.1.1 Inner-solver suboptimality (optimization error). For a fixed batch B and fixed parameters (θ, γ, β), we define the supremum objective function value when maximizing the function $L_{\text{in}}(B'; B, \theta, \gamma, \beta)$ with respect to the variable B' as follows:

$$\mathcal{L}^\star_{\text{in}}(B; \theta, \gamma, \beta) := \sup_{B'} \mathcal{L}_{\text{in}}(B'; B, \theta, \gamma, \beta). \tag{85}$$

That is, $\mathcal{L}^\star_{\text{in}}(B; \theta, \gamma, \beta)$ means the largest objective function value that $\mathcal{L}_{\text{in}}(B'; B, \theta, \gamma, \beta)$ can take when varying B', while fixing B and (θ, γ, β).

In implementation, we do not exactly find B' that achieves the above supremum, but instead use the result of performing J-step gradient ascent on B', denoted as B'_J. Here, for the same B and same (θ, γ, β), we define the difference between (i) the supremum objective function value $\mathcal{L}^\star_{\text{in}}(B; \theta, \gamma, \beta)$ and (ii) the objective function value $\mathcal{L}_{\text{in}}(B'_J; B, \theta, \gamma, \beta)$ at the J-step result B'_J as the optimization error, as follows:

$$\varepsilon_{\text{opt}}(B, J; \theta, \gamma, \beta) := \mathcal{L}^\star_{\text{in}}(B; \theta, \gamma, \beta) - \mathcal{L}_{\text{in}}(B'_J; B, \theta, \gamma, \beta) \ \geq 0. \tag{86}$$

Therefore, $\varepsilon_{\text{opt}}(B, J; \theta, \gamma, \beta)$ quantifies the gap between the objective function value when exactly solving the batch-based inner maximization problem for batch B, and the objective function value when approximately solving the same problem with J-step gradient ascent.

D.1.2 Mini-batch sampling variability (stochastic estimation error). Next, we quantify the phenomenon where $\mathcal{L}_{\text{in}}(B'_J; B, \theta, \gamma, \beta)$ varies from batch to batch since the batch B is randomly sampled from the empirical distribution $\hat{P}_n$. To this end, fixing given (θ, γ, β) and given inner iteration budget J, we define the expected value of the batch-based value $\mathcal{L}_{\text{in}}(B'_J; B, \theta, \gamma, \beta)$ calculated for a random mini-batch B as follows:

$$\bar{\mathcal{L}}_{\text{in},J}(\theta, \gamma, \beta) := \mathbb{E}\big[\, \mathcal{L}_{\text{in}}(B'_J; B, \theta, \gamma, \beta) \,\big]. \tag{87}$$

Here, the expectation $\mathbb{E}[\cdot]$ is taken over (i) the random sampling of the mini-batch B, and (ii) any additional randomness if the inner ascent employs it (e.g., random initialization).

Now, we define the stochastic estimation error as follows, measuring how much the observed value in a specific mini-batch B deviates from the above expected value:

$$\varepsilon_{\text{stoch}}(B, J; \theta, \gamma, \beta) := \Big| \bar{\mathcal{L}}_{\text{in},J}(\theta, \gamma, \beta) - \mathcal{L}_{\text{in}}(B'_J; B, \theta, \gamma, \beta) \Big|. \tag{88}$$

Therefore, $\varepsilon_{\text{stoch}}(B, J; \theta, \gamma, \beta)$ represents the degree to which the batch-based inner objective value can fluctuate depending on the mini-batch selection, even under the same (θ, γ, β) and same J.

In Supplementary Sec. D.2, we present the procedure for actually estimating ε_{opt} and $\varepsilon_{\text{stoch}}$ during the training process (experimental setup, validity discussion), along with the observed results and interpretations according to changes in settings (n_B, J, etc.).

D.2 Empirical Analysis of Approximation Gaps

In this section, we empirically measure the magnitudes of the two errors defined in Supplementary Sec. D.1.1–D.1.2 during the training process: (i) the optimization error ε_{opt} arising from the finite inner ascent step count J, and (ii) the stochastic estimation error $\varepsilon_{\text{stoch}}$ arising from mini-batch sampling. The purpose of this measurement is to verify at what level the discrepancy actually remains between the analysis based on the inner supremum defined at the distribution level and the mini-batch-based stochastic optimization performed in the actual implementation.

Evaluation protocol. At specific points during the training process (e.g., at regular training step intervals), we fix the outer variables (θ, γ, β) and sample M mutually independent mini-batches $B_1, \ldots, B_M$ from the empirical distribution $\hat{P}_n$. For each mini-batch B_m, we perform the procedure of maximizing the batch-level inner objective $\mathcal{L}_{\text{in}}(\cdot; B_m, \theta, \gamma, \beta)$ defined in Supplementary Sec. D.1.1 with respect to B' using two iteration budgets: (1) the J-step gradient ascent used in actual training, and (2) a stronger inner solver with a larger step budget J_{str} ($J_{\text{str}} = 4 \times J$). Here, for the gradient ascent, we keep the step size and update rule the same, and only increase the iteration budget.

Also, since the objective may not always monotonically increase depending on the step size in gradient ascent, we do not use a single final iterate for each budget, but instead use the iterate that yields the largest inner objective value in that trajectory as the representative solution. Specifically, denoting the perturbed batch defined by the t-th inner ascent iterate as $B'_{t,m}$,

$$B'_{J,m} := \arg \max_{t \in \{0,1,\dots,J\}} \mathcal{L}_{\text{in}}(B'_{t,m}; B_m, \theta, \gamma, \beta),$$
$$B'_{J_{\text{str}},m} := \arg \max_{t \in \{0,1,\dots,J_{\text{str}}\}} \mathcal{L}_{\text{in}}(B'_{t,m}; B_m, \theta, \gamma, \beta). \tag{89}$$

By this definition, $\mathcal{L}_{\text{in}}(B'_{J_{\text{str}},m}; B_m, \theta, \gamma, \beta) \geq \mathcal{L}_{\text{in}}(B'_{J,m}; B_m, \theta, \gamma, \beta)$ holds for each m.

Practical estimator of the optimization gap. By definition, the optimization error $\varepsilon_{\text{opt}}(B, J; \theta, \gamma, \beta)$ involves $\sup_{B'} \mathcal{L}_{\text{in}}(B'; B, \theta, \gamma, \beta)$, so it is difficult to calculate directly in actual experiments. Therefore, we consider the value provided by the J_{str}-step solver as a tight lower bound for the batch-level supremum, and construct a practical estimator via the difference with the J-step solver.

First, we define the average inner objective values of the two solvers as follows:

$$\mathcal{L}_{\text{in}}^{J} := \frac{1}{M} \sum_{m=1}^{M} \mathcal{L}_{\text{in}}(B'_{J,m}; B_m, \theta, \gamma, \beta),$$
$$\mathcal{L}_{\text{in}}^{J_{\text{str}}} := \frac{1}{M} \sum_{m=1}^{M} \mathcal{L}_{\text{in}}(B'_{J_{\text{str}},m}; B_m, \theta, \gamma, \beta). \tag{90}$$

Next, to reduce scale dependency, we define the optimization gap estimator normalized in the form of suboptimality as follows:

$$\hat{\varepsilon}_{\text{opt}} := \frac{\mathcal{L}_{\text{in}}^{J_{\text{str}}} - \mathcal{L}_{\text{in}}^{J}}{\left|\mathcal{L}_{\text{in}}^{J_{\text{str}}}\right|}. \tag{91}$$

A smaller $\hat{\varepsilon}_{\text{opt}}$ implies that the J-step inner solver used in actual training reaches an objective function value closer to that reached by the stronger solver (J_{str}-step).

Practical estimator of the stochastic estimation gap. Next, the stochastic estimation error $\varepsilon_{\text{stoch}}(\theta, \gamma, \beta)$ represents how much the objective function $\mathcal{L}_{\text{in}}(B'_J; B, \theta, \gamma, \beta)$ observed in a single mini-batch can deviate from the expected value over mini-batch sampling. Since the expected value cannot be calculated directly, we use a proxy that summarizes the variability of observed values for $N_B = 16$ independent mini-batches at fixed (θ, γ, β).

First, taking $\mathcal{L}_{\text{in}}^{J}$ in Eq. (90) as the reference value, we define the normalized mean absolute deviation as follows:

$$\hat{\varepsilon}_{\text{stoch}} := \frac{1}{N_B} \sum_{m=1}^{N_B} \frac{\left|\mathcal{L}_{\text{in}}(B'_{J,m}; B_m, \theta, \gamma, \beta) - \mathcal{L}_{\text{in}}^{J}\right|}{\left|\mathcal{L}_{\text{in}}^{J}\right|}. \tag{92}$$

A larger $\hat{\varepsilon}_{\text{stoch}}$ implies that the observed value of the inner objective varies significantly depending on mini-batch selection even for the same (θ, γ, β), and consequently, the discrepancy between the analysis described at the distribution level and the implemented stochastic training may increase.

Results and interpretation. We measured $\hat{\varepsilon}_{\text{opt}}$ and $\hat{\varepsilon}_{\text{stoch}}$ during the training process on PACS, and summarized the final values (final) and the maximum values observed during training (max) in Table 4 and Figure 3. First, from an optimization perspective, the full variant showed the smallest values with $\hat{\varepsilon}_{\text{opt}}^{\text{final}} = 0.0039$ and $\hat{\varepsilon}_{\text{opt}}^{\text{max}} = 0.0078$, suggesting that the default J-step inner solver provides sufficiently tight values compared to the stronger solver (J_{str}-step). In contrast, the rand G_Y variant maintained large $\hat{\varepsilon}_{\text{opt}}$ throughout the training period (final 0.0183, max 0.0204), showing a tendency to approximate the batch-level inner objective more loosely under the same iteration budget. Also, the w/o D_G & G_Y variant had an intermediate final value, but a period where it rose to $\hat{\varepsilon}_{\text{opt}}^{\text{max}} = 0.0154$ was observed during training, indicating that the approximation quality of the inner problem optimization can degrade in certain training periods.

Table 4: Estimated errors arising from the inner maximization. Experiments are conducted on PACS.

Variant	$\hat{\varepsilon}_{\text{opt}}^{\text{final}}$	$\hat{\varepsilon}_{\text{opt}}^{\text{max}}$	$\hat{\varepsilon}_{\text{stoch}}^{\text{final}}$	$\hat{\varepsilon}_{\text{stoch}}^{\text{max}}$
rand G_Y	0.0183	0.0204	0.0062	0.0108
w/o D_G & G_Y	0.0062	0.0154	0.0106	0.0400
full	0.0039	0.0078	0.0063	0.0138

Table 5: Performance comparison under different optimization settings for analyzing optimization errors on PACS. We report domain-wise accuracy (%) when varying (a) the mini-batch size n_B and (b) the number of inner maximization steps J. P/A/C/S denote the Photo, Art painting, Cartoon, and Sketch domains, respectively.

(a) batch size

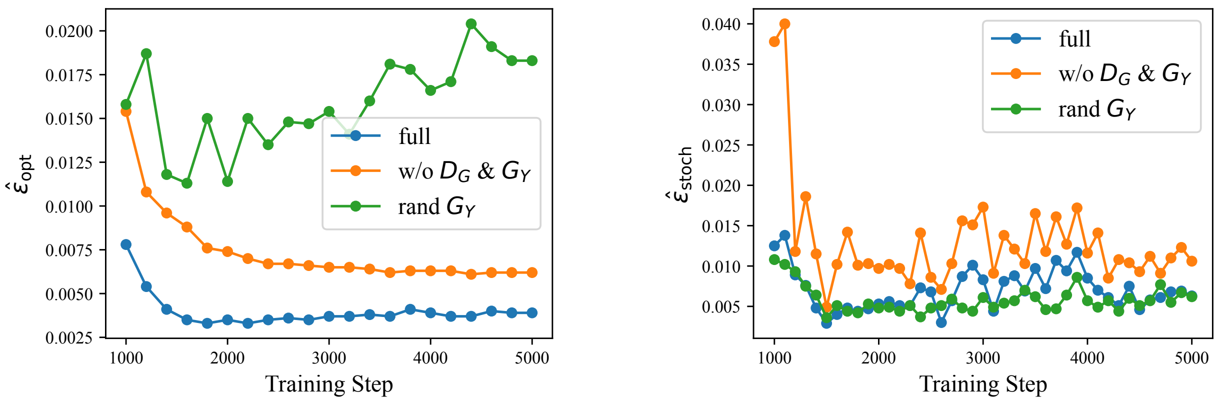
n_B	A	C	P	S	Avg.
8	88.5	80.0	97.9	77.9	86.1
16	91.2	80.1	98.1	79.7	87.3
32	90.3	84.7	98.4	82.3	88.9
64	90.6	83.5	98.5	81.4	88.5
128	90.3	82.3	98.7	81.0	88.1
256	91.2	81.6	98.5	80.6	88.0

(b) inner maximization steps

J	A	C	P	S	Avg.
5	91.4	82.0	98.7	79.8	88.0
10	90.1	83.2	98.8	80.8	88.2
15	90.3	84.7	98.4	82.3	88.9
20	90.0	82.8	98.8	79.8	87.8
25	90.7	83.9	98.6	81.3	88.6
30	90.0	82.9	98.8	81.4	88.3
35	91.4	82.3	98.6	80.4	88.2
40	90.5	83.8	98.7	81.0	88.5
45	91.1	83.7	98.5	81.5	88.7
50	90.1	82.9	98.2	80.9	88.0

From a stochastic perspective, the w/o D_G & G_Y variant showed a large spike with $\hat{\varepsilon}_{\text{stoch}}^{\text{max}} = 0.0400$ compared to other variants, implying that the variability of the observed inner objective values due to mini-batch sampling increased significantly. In contrast, the full variant showed relatively low and stable values with $\hat{\varepsilon}_{\text{stoch}}^{\text{final}} = 0.0063$ and $\hat{\varepsilon}_{\text{stoch}}^{\text{max}} = 0.0138$. In summary, the full variant tends to simultaneously lower (i) the approximation error of the J-step inner solver and (ii) the variability of mini-batch-based estimation, thereby mitigating the discrepancy between the distribution-level analysis and implementation.

Additional evidence from optimization settings (Table 5). To verify the relationship between $\hat{\varepsilon}_{\text{opt}}$ and $\hat{\varepsilon}_{\text{stoch}}$ and actual training performance, we report additional experimental results varying the optimization settings of CS-DRO on PACS in Table 5. Table 5 presents the average accuracy when varying (a) the mini-batch size n_B and (b) the number of inner maximization steps J. In Table 5(a), the average accuracy is highest at 88.9 when $n_B = 32$. Also, when n_B increases to 64, 128, and 256, the average accuracy is 88.5, 88.1, and 88.0, respectively, showing no significant improvement. When n_B is 8 and 16, the training performance is significantly degraded.



(a) $\hat{\varepsilon}_{\text{opt}}$

(b) $\hat{\varepsilon}_{\text{stoch}}$

Figure 3: Estimated errors arising from the inner maximization at each training step. Experiments are conducted on PACS.

This result supports the notion that as n_B gets smaller, the variability of $L_{\text{in}}(B'_J; B, \theta, \gamma, \beta)$ observed by mini-batch sampling may increase, and its impact can lead to degradation in generalization performance.

In Table 5(b), the average accuracy is 88.0 when $J = 5$, and is highest at 88.9 when $J = 15$. On the other hand, even if J is increased further, the average accuracy does not monotonically increase; for example, at $J = 20$, the average accuracy decreases to 87.8, and at $J = 45$, the average accuracy is 88.7, which is close to but does not exceed that of $J = 15$. Therefore, increasing J does not always guarantee performance improvement, and in actual training, there may exist a trade-off between the approximation precision of the inner solver (optimization error perspective) and training stability and computational cost. In summary, the results of Table 5 show that the choice of n_B and J in the mini-batch-based inner approximate optimization used by CS-DRO can have a substantial impact on the average accuracy, supporting that empirically checking the two approximation errors (optimization error and stochastic estimation error) defined in Supplementary Sec. D.1 within Supplementary Sec. D.2 is a meaningful diagnostic procedure.

E Impact of Structure Estimation Error

The preceding analysis premised a fixed proxy causal structure G. However, in practice, since we estimate G from finite samples, an error may occur between the empirically obtained $\hat{G}$ and the population-level structure $G^\star$. We investigate how conservatively this error makes the target inclusion certificate of Theorem 4.7, in the form of an upper bound on the error magnitude.

Let the optimal proxy minimizing the DAG learning objective (Eq. (14)) of Section 3.4 (main paper) be $G^\star \in \hat{\mathcal{G}}$. Here, $\hat{\mathcal{G}}$ is set as a compact feasible set defined by norm-bounded constraints. When defining the structural error as $\epsilon_G := \|\hat{G} - G^\star\|_F$, the error of $\hat{G}$ simultaneously perturbs the two constraints of the uncertainty set, $D_G(P, Q)$ and $W_G(P, Q)$. Under Assumption 4.1 and the compactness of $\hat{\mathcal{G}}$, since $D_G(P, Q)$ and $W_G(P, Q)$ satisfy Lipschitz continuity with respect to G respectively, we can obtain the following stability inequality regarding the structural error.

Lemma E.1 (Stability under structure error). *Assume Assumption 4.1 holds and $\hat{\mathcal{G}}$ is compact. Then, there exist constants $L_M, L_W < \infty$ such that for arbitrary $Q, P \in \mathcal{P}(\mathcal{V})$, the following holds:*

$$
\begin{aligned}
\left| D_{\hat{G}}(P, Q) - D_{G^\star}(P, Q) \right| &\le L_M \, \epsilon_G, \\
\left| W_{\hat{G}}(P, Q) - W_{G^\star}(P, Q) \right| &\le L_W \, \sqrt{\epsilon_G}.
\end{aligned} \tag{93}
$$

Provided that we set $\rho > \rho_0 + L_W \sqrt{\epsilon_G}$ and $\kappa > \kappa_0 + L_M \epsilon_G$ for a meaningful bound.

Proof. Let the structure estimation error be $\epsilon_G := \|\hat{G} - G^\star\|_F$. First, since $\hat{\mathcal{G}}$ is compact, the upper bound $\mathsf{B}_{\hat{\mathcal{G}}}$ on $\hat{\mathcal{G}}$ appears as follows:

$$
\mathsf{B}_{\hat{\mathcal{G}}} := \sup_{G \in \hat{\mathcal{G}}} \|G\|_F < \infty. \tag{94}
$$

Also, since $\mathcal{V} = \mathcal{Z} \times \mathcal{Y}$ is compact by Assumption 4.1, it appears as follows:

$$
\mathsf{B}_{\mathcal{V}} := \sup_{v \in \mathcal{V}} \|v\|_2 < \infty, \qquad \operatorname{diam}_2(\mathcal{Z}) := \sup_{z, z' \in \mathcal{Z}} \|z - z'\|_2 < \infty. \tag{95}
$$

Step 1. By Definition 3.3 and Eq. (8) of the main paper, $D_G(P, Q) = \|M_G(Q) - M_G(P)\|_F$. In Supplementary Sec. B.1, the entry-wise expectation representation of $M_G(\cdot)$ was given (see Lemma B.1 and Eq. (38)), and in particular, the pointwise gradient matrix at $A = \mathbf{1}$ appears as follows:

$$
\nabla_A \|r(v, A)\|_2^2 \Big|_{A=\mathbf{1}} = -2\Big(G \odot \big(r_G(v)\, v^\top\big)\Big), \qquad r_G(v) := v - G^\top v. \tag{96}
$$

Therefore, by Lemma B.1, it appears as follows:

$$
M_G(\nu) = \mathbb{E}_{v \sim \nu}\left[\nabla_A \|r(v, A)\|_2^2 \Big|_{A=\mathbf{1}} \right], \qquad \forall \nu \in \mathcal{P}(\mathcal{V}). \tag{97}
$$

Now, fix $v \in \mathcal{V}$ and let $\Delta G := G_1 - G_2$ for $G_1, G_2 \in \hat{\mathcal{G}}$. Since $r_{G_1}(v) - r_{G_2}(v) = -(G_1 - G_2)v = -\Delta G\, v$, using Eq. (96), it appears as follows:

$$
\begin{aligned}
&\Big\| -2\big(G_1 \odot (r_{G_1}(v) v^\top)\big) + 2\big(G_2 \odot (r_{G_2}(v) v^\top)\big) \Big\|_F \\
&\quad \le 2\|\Delta G\|_F \, \|r_{G_1}(v) v^\top\|_F + 2\|G_2\|_F \, \|(\Delta G\, v) v^\top\|_F.
\end{aligned} \tag{98}
$$

Here, $\|r_{G_1}(v)\, v^\top\|_F = \|r_{G_1}(v)\|_2\, \|v\|_2$, $\|(\Delta G\, v)\, v^\top\|_F = \|\Delta G\, v\|_2\, \|v\|_2$, and since $\|r_{G_1}(v)\|_2 = \|(I - G_1)v\|_2 \le (1 + \|G_1\|_F)\|v\|_2 \le (1 + \mathrm{B}_{\hat{\mathcal{G}}})\|v\|_2$, $\|\Delta G\, v\|_2 \le \|\Delta G\|_F \|v\|_2$, from Eq. (98), the following holds:

$$\begin{aligned}\Big\| \nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1},\,G=G_1} - \nabla_A \|r(v,A)\|_2^2\big|_{A=\mathbf{1},\,G=G_2}\Big\|_F \\ \le 2\big(1 + 2\mathrm{B}_{\hat{\mathcal{G}}}\big)\|v\|_2^2\, \|\Delta G\|_F.\end{aligned} \tag{99}$$

Substituting this into Eq. (97), taking the expectation, and using $\|v\|_2 \le \mathrm{B}_{\mathcal{V}}$, it appears as follows:

$$\|M_{G_1}(\nu) - M_{G_2}(\nu)\|_F \le 2\big(1 + 2\mathrm{B}_{\hat{\mathcal{G}}}\big)\mathrm{B}_{\mathcal{V}}^2\, \|G_1 - G_2\|_F, \quad \forall \nu \in \mathcal{P}(\mathcal{V}). \tag{100}$$

Now, applying the reverse triangle inequality and the triangle inequality to $D_G(P,Q) = \|M_G(Q) - M_G(P)\|_F$, it appears as follows:

$$\big|D_{G_1}(Q,P) - D_{G_2}(Q,P)\big| \le \|M_{G_1}(Q) - M_{G_2}(Q)\|_F \tag{101}$$

$$+ \|M_{G_1}(P) - M_{G_2}(P)\|_F. \tag{102}$$

Applying Eq. (100) to Eq. (102) and setting $G_1 = \hat{G}$, $G_2 = G^\star$, the following holds:

$$\big|D_{\hat{G}}(P,Q) - D_{G^\star}(P,Q)\big| \le 4\big(1 + 2\mathrm{B}_{\hat{\mathcal{G}}}\big)\mathrm{B}_{\mathcal{V}}^2\, \epsilon_G. \tag{103}$$

Therefore, the first inequality is obtained as $L_M := 4(1 + 2\mathrm{B}_{\hat{\mathcal{G}}})\mathrm{B}_{\mathcal{V}}^2$.

Step 2. In Eq. (9), c_G assigns an ∞ penalty to label mismatch. Thus, in the case where $W_G(P,Q) < \infty$ (the label-preserving setting considered in this paper), any coupling with finite expected cost must be supported on the set where $y = y'$. We now compare only the finite part $\|G_Y \odot (z - z')\|_2^2$ at $y = y'$.

(i) Lipschitz bound of G and G_Y. Letting $\mathrm{Se}_Y(\cdot)$ be the selection operator that extracts the d coordinates entering Y after selecting the parent dimension of Y and applying sigmoid, it appears as follows: $G_Y = \mathrm{sigmoid}\big(\mathrm{Se}_Y(G)\big)$. Here, sigmoid is globally 1/4-Lipschitz since $\sup_t |\mathrm{sig}'(t)| \le 1/4$, and since selection is coordinate extraction, $\|\mathrm{Se}_Y(G_1) - \mathrm{Se}_Y(G_2)\|_2 \le \|G_1 - G_2\|_F$ holds. Thus, it appears as follows:

$$\|\hat{G}_Y - G_Y^\star\|_2 \le \frac{1}{4}\, \|\hat{G} - G^\star\|_F = \frac{1}{4}\, \epsilon_G. \tag{104}$$

(ii) Uniform bound of cost perturbation. For arbitrary $z, z' \in \mathcal{Z}$, letting $\delta := z - z'$, the function $\psi(a) := \|a \odot \delta\|_2^2$ ($a \in [0,1]^d$) is differentiable and $\nabla_a \psi(a) = 2(a \odot \delta^{\odot 2})$, so $\|\nabla_a \psi(a)\|_2 \le 2\|\delta\|_2^2$ holds for $\|a\|_\infty \le 1$. Thus, by the mean value theorem, it appears as follows:

$$\big|\|\hat{G}_Y \odot \delta\|_2^2 - \|G_Y^\star \odot \delta\|_2^2\big| \le 2\|\delta\|_2^2\, \|\hat{G}_Y - G_Y^\star\|_2. \tag{105}$$

Also, since $\|\delta\|_2 \le \mathrm{diam}_2(\mathcal{Z})$, substituting Eq. (104) into Eq. (105), the following holds:

$$\sup_{z,z' \in \mathcal{Z}} \big|\|\hat{G}_Y \odot (z - z')\|_2^2 - \|G_Y^\star \odot (z - z')\|_2^2\big| \le \frac{1}{2}\, \mathrm{diam}_2(\mathcal{Z})^2\, \epsilon_G. \tag{106}$$

(iii) OT value comparison. Letting $\pi_{\hat{G}}$ be the optimal coupling for $c_{\hat{G}}$, the type-2 Wasserstein can be expressed as follows:

$$W_{G^\star}(P,Q)^2 = \inf_{\pi \in \Pi(Q,P)} \mathbb{E}_\pi\big[c_{G^\star}(v,v')\big] \le \mathbb{E}_{(v,v') \sim \pi_{\hat{G}}}\big[c_{G^\star}(v,v')\big]. \tag{107}$$

Also, since $\pi_{\hat{G}}$ is optimal for $c_{\hat{G}}$, $\mathbb{E}_{\pi_{\hat{G}}}[c_{\hat{G}}] = W_{\hat{G}}(P,Q)^2$. Therefore, the following holds:

$$\begin{aligned} W_{G^\star}(P,Q)^2 &\le \mathbb{E}_{\pi_{\hat{G}}}\big[c_{G^\star}(v,v')\big] = \mathbb{E}_{\pi_{\hat{G}}}\big[c_{\hat{G}}(v,v')\big] \\ &\quad + \mathbb{E}_{\pi_{\hat{G}}}\big[c_{G^\star}(v,v') - c_{\hat{G}}(v,v')\big] \\ &\le W_{\hat{G}}(P,Q)^2 \\ &\quad + \sup_{z,z' \in \mathcal{Z}} \big|\|\hat{G}_Y \odot (z - z')\|_2^2 - \|G_Y^\star \odot (z - z')\|_2^2\big|. \end{aligned} \tag{108}$$

Taking the square root of both sides and using $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$, we can obtain:

$$\begin{aligned} W_{G^\star}(P,Q) \le & W_{\hat{G}}(P,Q) \\ & + \left(\sup_{z,z' \in \mathcal{Z}} \big|\|\hat{G}_Y \odot (z - z')\|_2^2 - \|G_Y^\star \odot (z - z')\|_2^2\big| \right)^{1/2}. \end{aligned} \tag{109}$$

The reverse direction can be obtained identically:

$$\begin{aligned} \big|W_{\hat{G}}(P,Q) - & W_{G^\star}(P,Q)\big| \le \\ & \left(\sup_{z,z' \in \mathcal{Z}} \big|\|\hat{G}_Y \odot (z - z')\|_2^2 - \|G_Y^\star \odot (z - z')\|_2^2\big| \right)^{1/2}. \end{aligned} \tag{110}$$

Combining with Eq. (106), the following holds:

$$\big|W_{\hat{G}}(P,Q) - W_{G^\star}(P,Q)\big| \le \frac{1}{\sqrt{2}}\, \mathrm{diam}_2(\mathcal{Z})\, \sqrt{\epsilon_G}.$$

Therefore, the second inequality is obtained as $L_W := \frac{1}{\sqrt{2}}\, \mathrm{diam}_2(\mathcal{Z})$.

The L_M, L_W defined above are finite due to the compactness of $\hat{\mathcal{G}}$ and $\mathcal{V}$, and do not depend on P, Q. □

By Assumption 4.6, when the minimal radius and budget (ρ_0, κ_0) based on $G^\star$ exist, the radius and budget calculated based on $\hat{G}$ for the same target P^T conservatively inflate as follows:

$$W_{\hat{G}}(P, P^T) \le \rho_0 + L_W \sqrt{\epsilon_G}, \qquad D_{\hat{G}}(P, P^T) \le \kappa_0 + L_M \epsilon_G. \tag{111}$$

This inflation of (ρ, κ) makes the sufficient condition for guaranteeing target inclusion stricter. That is, when using $\hat{G}$, the residual margins for inclusion are tightened to $W_{\hat{G}}(P, \hat{P}_n) \le \rho - (\rho_0 + L_W \sqrt{\epsilon_G})$ and $D_{\hat{G}}(P, \hat{P}_n) \le \kappa - (\kappa_0 + L_M \epsilon_G)$. Consequently, the probabilistic lower bound presented in Eq. (23) (main paper) converges to a more conservative value in proportion to the structural error ϵ_G.

F Differences to Structural Causal OT

We compare with recent studies that integrate causal structure into OT, to discuss where CS-DRO enforces structural preservation and why it is more suitable for high-dimensional DG. Cheridito and Eckstein [6] and Ehyaei et al. [8] both, instead of using the entire coupling set $\Pi(P,Q)$ allowed in Wasserstein distance, restrict to the subset $\Pi_C(P,Q) \subset \Pi(P,Q)$ that satisfies a causal constraint C. Using this, OT is defined as above; specifically, the structure-constrained OT discrepancy is redefined as:

$$W_C(P,Q) := \inf_{\pi \in \Pi_C(P,Q)} \Big\{\mathbb{E}_{(v,v') \sim \pi}\big[c(v,v')\big]\Big\}^{1/2}. \tag{112}$$

In Cheridito and Eckstein [6] Π_G is defined to allow only couplings that are consistent with the information structure induced by a given DAG G (i.e. G-causal or G-bicausal couplings), aiming to leave only graph-consistent transport. In contrast, Ehyaei et al. [8] go beyond the DAG level by introducing F-compatible plans $\Pi_{\mathcal{F}}$ that preserve a structural equation models (SEMs) $\mathcal{F}$, thereby constructing a structural causal Wasserstein distance and the corresponding uncertainty set.

The key limitation of this line of approaches is that imposing C as a hard constraint on the feasible coupling set conflicts with DG learning. Ehyaei et al. [8], in order to preserve not only G but also the mechanism $\mathcal{F}$, rely on strong SEM-based assumptions (such as independent exogenous noises, counterfactual identifiability, and bijective reduced-form mappings), and their actual computation is carried out via a projection method using a bijective mapping function. Therefore, even if one learns an approximate graph $\hat{G}$ in the representation space and applies it, it is difficult to directly satisfy the plan-space definition required by Ehyaei et al. [8].

Meanwhile, the approach of Cheridito and Eckstein [6], although defined solely through a DAG, requires repeatedly solving a constrained OT subproblem over Π_G. This makes it difficult to obtain gradients for an end-to-end neural network training loss, so the method cannot be directly integrated into model training. Furthermore, [8] explicitly acknowledge the limitation that no computational method is provided to directly estimate a G-causal solution.

In contrast, CS-DRO is designed for DG in the following way: within a standard WDRO, it enforces structural preservation via the differentiable loss D_G, and through coupled constraint sets and Lagrangian relaxation it provides a tractable structure-guided DRO objective for high-dimensional DG.

G Implementation Details

G.1 Model Structure

In this study, we define the predictive model f_θ as $f_\theta = w \circ \phi$. Here, ϕ is a featurizer that extracts a representation z from an input image x, and w is a linear classifier that takes z as input and outputs class logits. In our experiments, we use a standard image classification backbone such as ResNet-50 for ϕ. Performing the DAG learning and DRO operations directly on the backbone representation $z \in \mathbb{R}^d$ can be computationally expensive and disadvantageous for optimization stability when d is large. Therefore, we add a linear projection head $\tilde{w}$ to map z into a lower-dimensional space and use a normalized reduced representation $\tilde{z} \in \mathbb{R}^{d'}$ (with $d' \ll d$):

$$\tilde{z} \ = \ \mathrm{LN}(\tilde{w}z + \tilde{b}) \in \mathbb{R}^{d'} \qquad (d' \ll d), \tag{113}$$

Here $\tilde{w} \in \mathbb{R}^{d' \times d}$ is the linear transformation, and $\mathrm{LN}(\cdot)$ denotes LayerNorm. All subsequent DAG learning, DRO, and final classification are performed based on $\tilde{z}$.

We define the variable vector (including the label node) as $v = [\tilde{z}; y] \in \mathbb{R}^{d'+1}$, and set the graph parameter $G \in \mathbb{R}^{(d'+1)\times(d'+1)}$ to be a trainable matrix. Here y is treated as a variable corresponding to a single label node. G operates on v in a linear structural equation form and is trained so that the first d' components of $G^\top v$ reconstruct $\tilde{z}$. Furthermore, to ensure that DAG learning does not stop at simple reconstruction but also captures the dependency in the direction $\tilde{z} \to y$, we introduce an auxiliary classifier g. Specifically, let the column of G that feeds into the label node y be $G_Y \in \mathbb{R}^{d'}$, and use it as a per-dimension weight on $\tilde{z}$ to predict the label as follows:

$$\hat{y} \ = \ g(G_Y \odot \tilde{z}), \tag{114}$$

where $\odot$ denotes element-wise product. This construction encourages G_Y to reflect the coordinates corresponding to the parents of the label node, helping G to better capture the $\tilde{z} \to y$ relationship.

The final prediction of f_θ is also made on the reduced representation $\tilde{z}$, using the form $w(G_Y \odot \tilde{z})$ to emphasize the coordinates important for label prediction. In doing so, to ensure that G_Y functions reliably as a set of per-dimension weights, we apply a median-based normalization and a sigmoid so that each element of G_Y is adjusted to have a value between 0 and 1:

$$\mathrm{mad} = \mathrm{median}\Big(\big|G_Y - \mathrm{median}(G_Y)\big|\Big),$$
$$G_Y = \mathrm{sigmoid}\left(\frac{G_Y - \mathrm{median}(G_Y)}{\mathrm{mad}}\right).$$

Here mad is the median absolute deviation of the components of G_Y, and $\mathrm{sigmoid}(\cdot)$ denotes the sigmoid function. This outlier-insensitive scaling mitigates the tendency of G_Y to become overly biased toward 0 or 1, helping it serve as a more stable set of per-dimension weights during training.

G.2 Training Procedure

Algorithm 1 (main paper) shows a summary of the training procedure.

G.2.1 Warm-up. In the warm-up stage, we repeatedly sample from the pooled source data $\cup_e \mathcal{D}^e$ and train (ϕ, w) via ERM. Since the initial z may have strong inter-dimensional dependencies or unstable variance, which could destabilize subsequent structure learning, we apply regularization to control the statistical properties of z in addition to the standard classification loss. Specifically, during the warm-up period (steps ≤ 300), we minimize the cross-entropy loss while additionally imposing: (i) an HSIC-based penalty to alleviate dependencies between different dimensions of z, and (ii) a variance stabilization penalty to prevent the variance of z from diminishing excessively.

The HSIC-based penalty aims to reduce the statistical dependence between any two distinct dimensions of the mini-batch representations. We compute an RBF-kernel HSIC value for a number of dimension pairs and average them to construct this penalty. For efficiency, we do not use all possible pairs of dimensions; instead, we compute HSIC on 64 randomly selected pairs of dimensions. The bandwidth of the RBF kernel is set to the median of the pairwise distances within the batch. The variance stabilization penalty is applied only when the variance of each dimension becomes too small (to prevent collapse of z). In implementation, we use a hinge-shaped penalty proportional to the shortfall whenever a dimension's standard deviation falls below 1. The weight of these regularization terms is increased linearly from 0.001 to 0.1 during the beginning of training so that classification performance is first stabilized, and the influence of the regularization gradually grows, avoiding overly strong constraints at the very start of warm-up. In addition, to compose reliable samples for proxy DAG learning, during warm-up we accumulate the mean and variance of z for each domain–label pair (e, c) and construct a Gaussian distribution $\mathcal{N}(\mu_{e,c}, \sigma_{e,c})$. These accumulated statistics act as a buffer so that even if the statistics of

z slowly change as the model trains, the samples used for structure learning do not fluctuate too drastically.

G.2.2 DAG learning. Next, the proxy DAG is trained using synthetic samples generated from the per-domain-per-class Gaussian distributions constructed above. In implementation, after the Gaussians $\mathcal{N}(\mu_{e,c}, \sigma_{e,c})$ have been sufficiently populated in the later warm-up phase (100 < step < 300), at each training step we perform 3 repeated updates of the DAG parameters to adequately progress the structure learning. In each update, we optimize two losses: (i) a reconstruction loss that ensures $\tilde{z}$ is linearly reconstructed for the input vector including the label node, and (ii) an auxiliary classification loss obtained by feeding the element-wise product $G_Y \odot \tilde{z}$ into the auxiliary classifier g to predict the label. We minimize this auxiliary classification loss alongside the reconstruction loss. This way, structure learning is encouraged not to remain a mere reconstruction task but to also reflect the dependency in the $\tilde{z} \rightarrow y$ direction. We also gradually enforce the acyclicity constraint $h(G)$ (to make G a DAG) via an augmented Lagrangian approach. In practice, although in the main text $h(G)$ is added to the objective with weight λ_a, following prior works [11, 30, 31], we include it in the objective in the form $\lambda'_a h(G)^2 + \lambda_a h(G)$, where λ_a and λ'_a are dual variables initially set to 0 and 1, respectively, and increased as $h(\cdot)$ approaches convergence. This strategy allows the model to explore slight constraint violations freely in the early stages while increasingly enforcing the DAG constraint later. Finally, we add an $\mathcal{L}_1$ regularization term to the objective with weight $\lambda_{l_1} = 0.01$ to suppress unnecessary edges. Once the warm-up stage is completed, we fix G and G_Y, and update only (ϕ, w) via CS-DRO going forward.

G.2.3 Inner Maximization. With G fixed, the CS-DRO training proceeds as follows in each iteration. First, we sample a batch $B = \{(x_i, y_i)\}_{i=1}^{n_B}$ from all source domains and obtain $z_i = \phi(x_i)$ for each sample. In the inner step, starting from each z_i, we construct an adversarial representation z'_i by updating z' in the direction that increases the loss. We repeat this update J times to produce an augmented batch B'. Specifically, we initialize $z'_i = z_i + \mathrm{Uniform}(-10^{-4}, 10^{-4})$. We then compute the loss on the entire batch and perform $J = 15$ iterations of gradient ascent on z' (using Adam with step size $\eta' = 0.05$) for the same batch to obtain the adversarial batch B'.

G.2.4 Outer Minimization. After obtaining B' from the inner step, the outer step updates θ by minimizing the sample objective on this augmented batch. In addition, since ϕ is being updated during training while G is fixed, we add a structure-preserving loss $\mathcal{L}_G(B)$ (for batch B) to the objective with weight λ_G so that the current z does not deviate too far from the structure defined by G. For stable training, we also employ a logit-based regularization term $\mathcal{L}_{\mathrm{reg}}$, which uses a temperature τ and a bidirectional KL divergence (with a stop-gradient) to mitigate overly sharpened predictions. The dual variables (γ, β) are updated by projected gradient steps based on the amounts of constraint violation. Specifically, as given in Algorithm 1 (main paper), we update: $\gamma \leftarrow \gamma + \eta_\rho\,[\mathbb{E}_B[c(z, z')] - \rho^2]_+$, $\beta \leftarrow \beta + \eta_\kappa\,[D_G(B', B) - \kappa]_+$, ensuring that γ and β remain non-negative after each update.

G.2.5 Radius Update. Finally, the radius parameter ρ is estimated during training based on the current adversarial radius in the representation space. Specifically, every 100 steps we obtain the representations z and labels y for the training dataset, and for each sample we solve a closest misclassification problem to find the nearest z'' that causes a misclassification. Here z'' is found by an optimization using the Carlini–Wagner (CW) method [5]. We then compute the distance $\|z'' - z\|_2$ for each sample, average these distances, and use the result as the estimate $\hat{\rho}^2$ at the current step. For numerical stability, the computed $\hat{\rho}^2$ is accumulated via an exponential moving average and we update $\rho^2 \leftarrow 0.9\,\rho^2 + 0.1\,\hat{\rho}^2$.

Table 6: Hyperparameter values used for each dataset.

Hyperparameter	PACS	VLCS	OfficeHome	TerraInc.
learning rate	5-e5	5-e5	5-e5	5-e5
weight decay	1-e6	1-e6	1-e6	1-e6
max step	5000	5000	5000	5000
frequency of evaluation	200	200	200	200
κ	10^0	10^0	10^{-3}	10^{-1}
λ_G	10^1	10^1	10^0	10^1

G.3 Detailed Hyperparameters

For reproducibility, we summarize all hyperparameters used throughout the paper and their values. Unless stated otherwise, the settings below were fixed to the same values for all datasets.

DomainBed optimization and evaluation settings. Following the DomainBed protocol, we split the source domain data into train / validation with an 8:2 ratio, and measured performance using a leave-one-domain-out evaluation (rotating which domain is held out as target). All experiments were run 3 times with different random seeds. We used an ImageNet-pretrained ResNet-50 as the backbone and Adam as the optimizer. The learning rate was 5×10^{-5}, weight decay 10^{-6}, and training ran for a total of 5,000 steps. Evaluation was performed every 200 steps. Each mini-batch was constructed to include samples from all source domains (we used 32 samples per domain per batch).

Reduced representation and G_Y normalization. We applied a projection head and LayerNorm to the backbone output z to define the reduced representation as $\tilde{z} = \mathrm{LN}(\tilde{W}z + \tilde{b})$. The reduced dimension was fixed to $d' = 256$. We also stabilized G_Y by applying median and MAD-based normalization followed by a sigmoid (as described in Section G.1).

Warm-up and statistics accumulation. The warm-up phase was fixed to end at step 300. The HSIC penalty was computed on 64 randomly selected dimension pairs (instead of all pairs), and the RBF kernel bandwidth was set to the median of pairwise distances in the batch. The variance stabilization penalty was implemented in hinge form (penalizing proportionally to the shortfall when a dimension's standard deviation dropped below 1), and its weight was linearly increased from 0.001 to 0.1 over the warm-up. Also during warm-up, for each domain–label pair (e, c) we accumulated $(\mu_{e,c}, \sigma_{e,c})$ and constructed a Gaussian distribution $\mathcal{N}(\mu_{e,c}, \sigma_{e,c})$. We drew 200 samples per class per domain from this Gaussian for DAG learning.

Proxy DAG learning. Proxy DAG learning was performed during the interval $100 < \text{step} < 300$. In this interval, at each training step the DAG parameter update was repeated 3 times. The acyclicity constraint was handled via an augmented Lagrangian; we set the initial values to $\lambda_a = 0$, $\lambda'_a = 1$ [11, 30]. The weight for l_1 regularization (to encourage sparsity) was set to $\lambda_{l_1} = 0.01$. Other implementation details of DAG learning followed the default settings of [11].

CS-DRO inner maximization. The adversarial representation was initialized as $z'_i = z_i + \text{Uniform}(-10^{-4}, 10^{-4})$. The number of inner iterations was fixed to $J = 15$, with Adam as the inner optimizer and step size $\eta' = 0.05$.

CS-DRO outer settings and regularization. The logit regularization term $\mathcal{L}_{\text{reg}}$ used temperature $\tau = 10$ and weight $\lambda_r = 1$. For the projected updates of the dual variables, we used the step sizes (η_ρ, η_κ) as defined in Algorithm 1 (main paper), and fixed both η_ρ and η_κ to 10^{-4} in all experiments. The weight λ_G of the structure-preserving loss and the budget κ of the structure-preserving constraint were tuned per dataset, while all other outer-step settings were identical across datasets.

Periodic radius estimation and moving average. Every 100 steps, ρ was re-estimated via the closest misclassification approach to obtain $\hat{\rho}^2$ (using the CW method), and the moving average update $\rho^2 \leftarrow 0.9\,\rho^2 + 0.1\,\hat{\rho}^2$ was applied.

Dataset-specific tuning of λ_G and κ. Following Qiao and Peng [23], we chose λ_G and κ from the log-scale set $\{10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}\}$ based on validation performance. Ultimately, for PACS and VLCS we used $(\kappa, \lambda_G) = (10^0, 10^1)$, for OfficeHome $(10^{-3}, 10^0)$, and for TerraIncognita $(10^{-1}, 10^1)$. Table 6 summarizes these choices.

H Additional Experiments

H.1 Large Dataset

We evaluate four datasets in Table 1 (main paper). These include PACS [14] (9,991 images, 7 classes, 4 domains), VLCS [9] (10,729 images, 5 classes, 4 domains), OfficeHome [26] (15,588 images, 65 classes, 4 domains), and TerraIncognita [2] (24,788 images, 10 classes, 4 domains). While these datasets are standard in DG research, their relatively limited number of classes and domain diversity necessitate further validation of the proposed method under larger-scale and more complex distribution shifts. To address this, we conduct additional experiments on DomainNet [20], which is a significantly larger and more complex dataset. DomainNet consists of 586,575 images across 345 classes and 6 domains. It represents a much more challenging setup for DG due to the substantial visual style differences between domains and the high number of classes. Table 7 presents the results of the leave-one-domain-out evaluation on DomainNet. For this experiment, we set the total training steps to 15,000 and the evaluation frequency to 500, with $\kappa = 10^{-2}$ and $\lambda_G = 10^0$. Experimental results show that CS-DRO achieves an average accuracy of 47.8%, the highest among all compared methods. In particular, our method demonstrates consistent performance improvements over strong baselines such as GGA and SMIDG across multiple target domains, including Clipart (C), Infograph (I), Painting (P), Real (R), and Sketch (S). These results confirm that our design of restricting the uncertainty set in a structure-preserving manner operates effectively even on large-scale data. Although CS-DRO shows a slightly lower result compared to the state-of-the-art on Quickdraw (Q), it remains superior in terms of the overall mean accuracy. This suggests that CS-DRO successfully enhances general robustness in large-scale environments characterized by diverse stylistic variations.

Table 7: Performance comparison of the proposed method with existing methods on the DomainNet dataset. Results are reported as accuracy (%), averaged over three runs with different random seeds. Except for our method, all results are taken from the original literature. C, I, P, Q, R, and S denote the Clipart, Infograph, Painting, Quickdraw, Real, and Sketch domains, respectively.

Method	C	I	P	Q	R	S	Avg.
VREx [13]	47.3	16.0	35.8	10.9	49.6	42.0	33.6
RSC [10]	55.0	18.3	44.4	12.2	55.7	47.8	38.9
GroupDRO [24]	47.2	17.5	33.8	9.3	51.6	40.1	33.3
MTL [4]	57.9	18.5	46.0	12.5	59.5	49.2	40.6
Mixstyle [32]	51.9	13.3	37.0	12.3	46.1	43.4	34.0
SagNet [19]	57.7	19.0	45.3	12.7	58.1	48.8	40.3
SAGM [27]	64.9	21.1	51.5	**14.8**	64.1	53.6	45.0
SMIDG [28]	64.2	23.6	53.5	14.5	62.6	54.7	45.5
GGA [1]	64.0	22.2	51.7	14.3	64.1	54.7	45.6
CS-DRO	**66.6**	**25.6**	**56.3**	13.4	**67.9**	**56.9**	**47.8**

H.2 DAG Learning Validity Verification

In this section, we analyze whether the proxy DAG is actually being effectively utilized in the proposed method, and how the quality of DAG learning affects the final DG performance. Since CS-DRO constructs the structure-preserving constraint and anisotropic cost based on the estimated DAG G and the label-related mask G_Y, if DAG learning is unstable or the structure is misestimated, the uncertainty set may be formed in an unrealistic way, degrading generalization performance. To verify this, we conduct experiments on: (i) the difficulty of DAG learning and performance as the reduced representation dimension d' changes, (ii) the stability of DAG estimation depending on the warm-up end step, and (iii) an ablation study on the usage of G and G_Y and the components of the DAG objective. The results are presented in Table 8 and Figure 4.

H.2.1 Effect of Representation Dimension on d'. DAG learning becomes more difficult to optimize as the number of variables to learn increases, and the reconstruction error grows or the influence of the acyclicity and sparsity constraints can change. Therefore, we use a reduced representation $\tilde{z} \in \mathbb{R}^{d'}$ obtained by mapping the representation z into a lower dimension, and observe the performance and DAG learning metrics while varying d'. In Table 8a, as d' decreases, the average performance drops, which is consistent with the effect of reduced discriminative power due to limited expressiveness from dimensionality reduction. On the other hand, when d' becomes excessively large, performance also drops again;

Table 8: Analysis of Structural Validity. We report the domain-wise accuracy (%) on the PACS dataset across (a) varying levels of latent dimensionality reduction and (b) different warm-up durations, alongside extended ablation studies.

(a) The reduced dimension d' of the latent feature vector z.

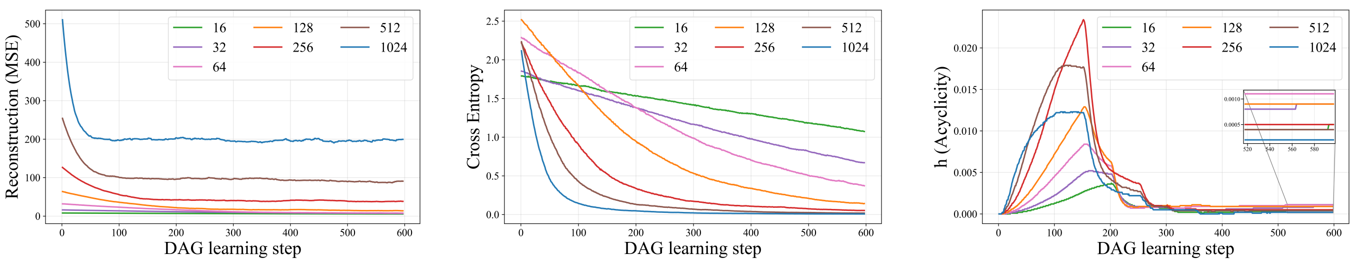
d'	A	C	P	S	Avg.
16	88.8	78.2	98.1	77.9	85.8
32	88.0	81.1	97.1	80.2	86.6
64	89.3	82.4	98.1	78.7	87.2
128	89.9	82.6	98.6	79.2	87.6
256	90.3	84.7	98.4	82.3	88.9
512	89.5	83.4	98.5	80.6	88.0
1024	88.7	84.1	98.0	79.2	87.5

(b) Warm-up end step

End Step	A	C	P	S	Avg.
50	88.2	82.7	98.6	79.1	87.1
100	88.8	82.5	98.7	81.3	87.8
150	90.2	83.6	98.9	80.6	88.3
200	89.5	83.9	98.4	82.6	88.6
250	90.6	82.4	98.4	82.1	88.4
300	90.3	84.7	98.4	82.3	88.9

(c) Extended Ablation Analysis

Variant	A	C	P	S	Avg.
w/o HSIC	89.7	83.3	98.2	82.6	88.5
w/o G_Y in w	90.1	83.3	97.6	81.4	88.1
Random G	88.8	82.1	97.3	80.5	87.2
Random G_Y	88.6	81.4	96.9	80.8	86.9
w/o $\mathcal{L}_1$	89.8	83.5	97.6	81.8	88.2
w/o h	89.3	82.5	98.2	80.1	87.5
CS-DRO	90.3	84.7	98.4	82.3	88.9



(a) Reconstruction **(b) Classification** **(c) Acyclicity**

Figure 4: Performance results across different latent dimensions relative to DAG learning steps.

in this case, DAG learning itself becomes difficult and the quality of the proxy structure deteriorates as well.

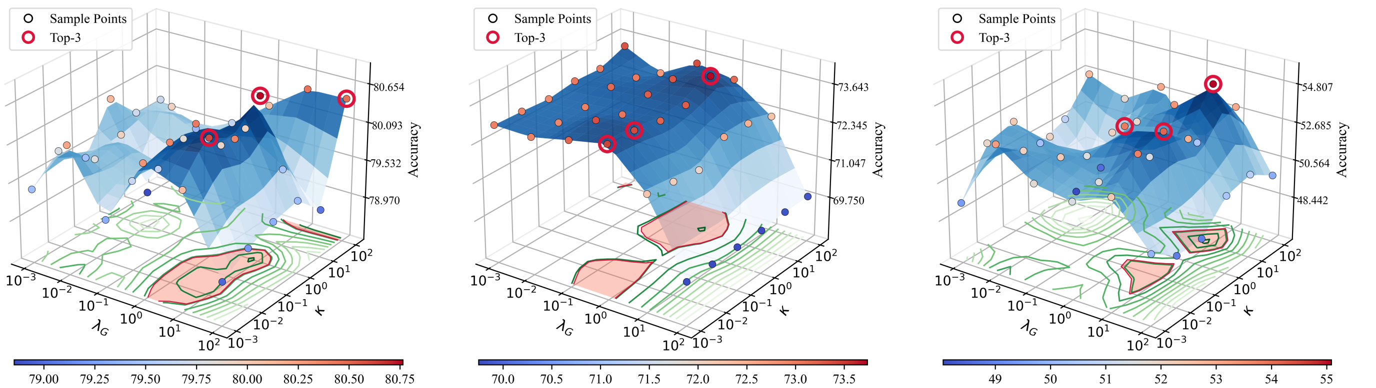
To see this more directly, in Figure 4 we compare the components of the DAG learning objective over training steps. First, in Figure 4a, as d' increases, the reconstruction error tends to converge at a higher value, which means that within the same training steps, G remains unable to sufficiently explain the reduced representation. In contrast, the cross-entropy in Figure 4b converges faster for larger d', which suggests that even if g (the label predictor) is relatively easy to optimize from the label prediction perspective, if the reconstruction of z is not adequately achieved, it becomes difficult to obtain a useful proxy DAG in the end.

Lastly, Figure 4c shows that $h(\cdot)$ decreases and converges to satisfy the DAG constraint within 300 steps in all settings. Notably, at $d' = 256$, the peak of $h(\cdot)$ appears the largest, which suggests that the model undergoes a pronounced process where it temporarily learns complex interactions including cycles to reconstruct the data, and then converges to an acyclic structure due to the penalty. In contrast, when d' is too small or too large, the peak is low, which indicates a high possibility that the learned structure cannot adequately explain the data—either due to early convergence to a local minimum for reconstruction or instability in learning—depending on the case.

In summary, when d' is too small, loss of representation power dominates, and when d' is too large, DAG learning (especially the reconstruction) becomes unstable, so the structure-based constraint cannot function effectively, resulting in performance degradation. In our experiments, this trade-off is best balanced at $d' = 256$, achieving the highest performance.

H.2.2 Effect of Warmup End Step. Next, we evaluate the effect of the timing of ending the warm-up (end step) on DAG estimation stability and DG performance (Table 8b). The warm-up stage serves to (i) alleviate inter-dimensional dependence via an HSIC-based regularizer, and (ii) build a batch of cumulative statistics so that the DAG can be learned based on more consistent information from the representation. In Table 8b, when the end step is short (50 to 100), performance is lower, which suggests that if the proxy DAG is estimated before the representation has sufficiently stabilized, the structure estimation error can be large. On the other hand, extending the end step to 150 or more generally improves performance, and performance remains relatively stable in the range of 150 to 300. This means that once the warm-up is secured to a certain extent, the distribution of input samples used for DAG learning becomes stable and, as sufficient DAG learning is performed, the structure-based constraint can restrict the uncertainty set more consistently.

H.2.3 Ablation Study on Proxy DAG Usage. Lastly, to see whether the learned DAG is actually used effectively in the proposed method, we conduct an additional ablation study as shown in Table 8c. First, "w/o HSIC" removes the inter-dimensional dependence alleviation in the warm-up, and the average performance decreases. This shows that if the formation of representations as independent causal variables is weakened, the stability of DAG learning can deteriorate. Next, "w/o G_Y in w" removes the enforcement of label-related dimensions via G_Y in the prediction process of f_θ, and the average



(a) VLCS (b) OfficeHome (c) TerraIncognita

Figure 5: Additional performance comparison results with respect to variations in κ and λ_G.

performance drops. This suggests that G_Y is not a mere add-on but plays an important role in forming a structurally aligned learning signal.

In particular, “Random G” and “Random G_Y” result in a large performance drop, meaning that if the structure-preserving constraint and anisotropic cost are defined in arbitrary directions, the uncertainty set fails to capture shifts compatible with the data-generating mechanism. Rather, it may allow unrealistic perturbations or suppress important feature directions, thereby hindering generalization. Lastly, even when the sparsity and acyclicity constraints are removed from the DAG objective ("w/o $\mathcal{L}_1$", "w/o h"), performance decreases. This is consistent with the tendency that without these constraint terms, G becomes overly dense or does not sufficiently satisfy the DAG property, weakening its role as a proxy structure and consequently reducing the effectiveness of the structure-preserving design.

H.2.4 Summary. The results in Table 8 and Figure 4 show that (i) when the proxy DAG is learned stably beyond a certain level, and (ii) when the learned G and G_Y are utilized to restrict the uncertainty set according to the proposed structure-preserving design, CS-DRO consistently exhibits improved generalization performance.

H.3 Additional Structural Control Analysis

We compared performance while varying κ and λ_G on VLCS, OfficeHome, and TerraIncognita. Figure 5 shows the leave-one-domain-out average accuracy (%) for combinations of (κ, λ_G) on each dataset. This experiment is an extension of Figure 2 (main paper) in the main text, not limited to PACS, and aims to verify whether the effect of the structure-preserving constraint is observed consistently across different datasets.

In the case of VLCS, relatively high performance appears when $\lambda_G \in \{10^0, 10^1\}$, and at those values a wide range of $\kappa \in \{10^{-3}, 10^{-2}, 10^{-1}, 10^0\}$ yields top performance combinations. In other words, under the setting that encourages z to sufficiently reflect the estimated G via L_G, the D_G-based constraint works stably and provides consistent gains across various κ values.

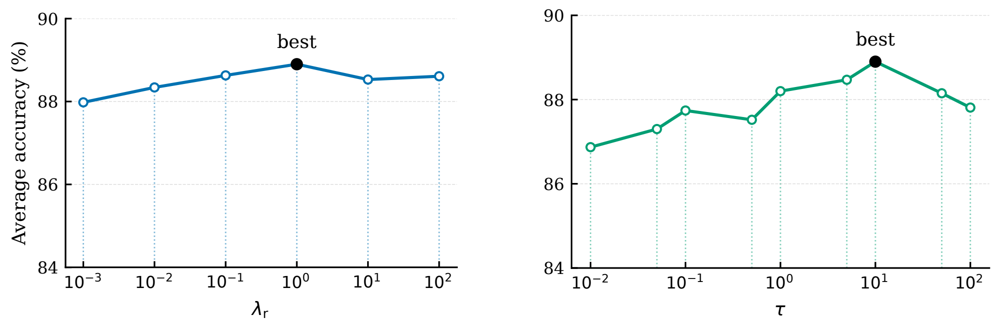
In OfficeHome, relatively superior results are observed at $\lambda_G \in \{10^{-1}, 10^0\}$, while performance changes little across the range of κ values. This indicates that, in this dataset, once a representation that follows the structure is secured, adjusting κ does not significantly sway the results.

In TerraIncognita, the top performance is also achieved at $\lambda_G \in \{10^0, 10^1\}$, and additionally the best range for κ is observed at $\{10^{-2}, 10^{-1}, 10^1, 10^2\}$. TerraIncognita has a large domain shift, so the performance surface fluctuates more, but nevertheless the fact that consistently top combinations appear when λ_G is sufficiently large supports that making z reflect the structure of G is crucial in the uncertainty set design.

In summary, across multiple benchmarks, high performance commonly emerges when λ_G is above a certain level, and the influence of κ either varies by dataset or is robust to changes in its value. This indicates that the structure-preserving constraint is consistently effective even under diverse domain shifts. In particular, it suggests that aligning the evolving representation with G is necessary to achieve high performance, and once this alignment is in place, the effect of κ in the structure-preserving constraint becomes more pronounced.

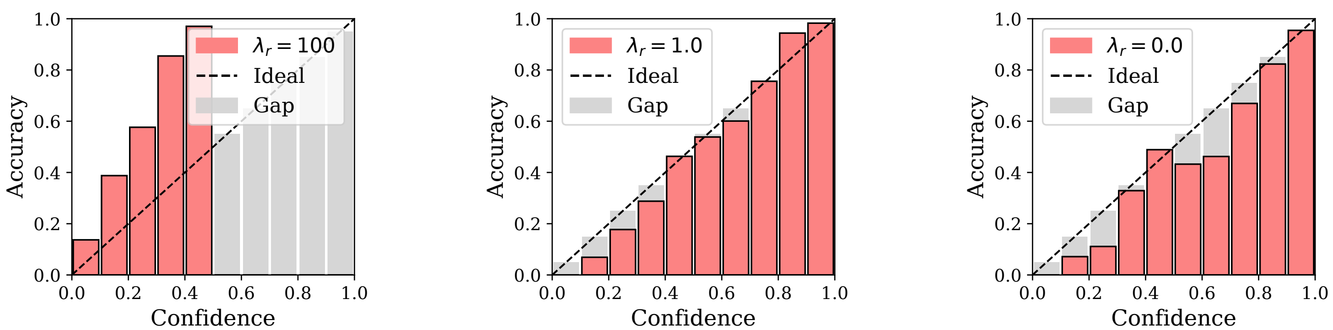
H.4 Effect of the Regularization Loss for Inducing Predictive Consistency

$\mathcal{L}_{\text{reg}}$ is a logit-based regularization term to control the prediction drift between the original representation z_i and a perturbed representation z_i'. It uses a temperature τ to smooth the prediction distribution (with the smoothed predictions used as a stop-gradient target) and is designed to minimize the bidirectional KL divergence (Eq. (17)-(18) of the main paper). We performed a hyperparameter analysis on τ and λ_r to examine how sensitively this design is reflected in actual performance. In the default setting we use $\tau = 10$ and $\lambda_r = 1$. For the analysis, we varied $\lambda_r \in \{10^{-3}, 10^{-2}, 10^{-1}, 10^0, 10^1, 10^2\}$ and $\tau \in \{10^{-2}, 10^{-1}, 10^0, 10^1, 10^2\}$ on a log scale, measuring the average accuracy on PACS. The results are presented in Figure 6. In Figure 6, even when λ_r is varied over a wide range, the performance changes are relatively gradual, and a certain level of robustness is observed with respect to the choice of the weight of $\mathcal{L}_{\text{reg}}$ and leading to a significant performance drop. Moreover, when τ is extremely large (e.g. $\tau = 10^2$), the



(a) Hyperparameter analysis (λ_r) **(b) Hyperparameter analysis (τ)**

Figure 6: Performance comparison on the PACS dataset under varying hyperparameter values of τ and λ_r associated with $\mathcal{L}_{\text{reg}}$.



(a) **(b)** **(c)**

Figure 7: Comparison of confidence calibration on the OfficeHome dataset as λ_G varies.

Table 9: Training time per step and total training time for PACS on a single NVIDIA RTX 3090. Wall-clock time includes evaluation overhead; results are averaged over three runs with different random seeds (mean ± std).

Method	Time/step (s)	Total (5000 steps) (s)
SAL†	0.26810 ± 0.00002	1340.49
DR-DG†	0.19984 ± 0.00013	999.18
GLOT-DR†	0.25961 ± 0.00027	1298.06
TRO†	0.20082 ± 0.00004	1004.07
MEDRO†	0.20945 ± 0.00001	1047.23
SR-WDRO†	0.20533 ± 0.00016	1026.63
CS-DRO	0.22622 ± 0.00031	1131.11

prediction distribution becomes overly flat and less discriminative, which likewise results in performance degradation. In summary, it is important to choose τ in a moderate range that mitigates excessive sharpening while maintaining discriminability; in our setting, the most stable performance is observed in the range of 10^0–10^1. On the other hand, in Figure 6b, τ shows a more direct effect on performance. For $\tau \leq 10^0$, applying a temperature actually causes predictions to become sharper (rather than smoother), weakening the stabilizing effect intended by $\mathcal{L}_{\text{reg}}$.

We further evaluated model calibration using reliability diagrams. Specifically, we divided the predictions into 10 bins by confidence level and computed the mean accuracy in each bin. An optimally calibrated model should show a diagonal pattern in the reliability diagram, which we plot as an ideal reference. In this analysis, we fixed τ at 10 and varied $\lambda_r \in \{0.0, 1.0, 100.0\}$ and varied $\mathcal{L}_{\text{reg}}$ to adjust the strength of $\mathcal{L}_{\text{reg}}$. In Figure 7c, when $\lambda_r = 0.0$, there exist ranges where the average accuracy per bin is higher or lower than the ideal diagonal, and overall the curve deviates from a consistent diagonal pattern. In contrast, Figure 7b with $\lambda_r = 1.0$ yields a curve closest to the ideal diagonal, and overall a stable calibration is observed. This suggests that $\mathcal{L}_{\text{reg}}$ not only suppresses the prediction drift between z_i and z_i', but the temperature-based smoothing also alleviates excessive sharpening, leading to more reliable probability estimates. Meanwhile, in Figure 7a, using an overly large weight ($\lambda_r = 100.0$) causes the prediction confidences to concentrate around roughly 0.4, yielding a narrowly skewed distribution. This can be interpreted as a case where the regularization is working too strongly, making the predictions overly smooth and consequently causing the probability outputs to converge to extremely conservative values. In summary, $\mathcal{L}_{\text{reg}}$ at an appropriate strength provides calibration close to the ideal diagonal pattern, but excessive regularization shrinks the confidence to a narrow range and can impair discriminability.

Table 10: Domain-wise performance on DomainBed. Results are reported as accuracy (%), averaged over three runs with different random seeds. For PACS, P/A/C/S denote the Photo, Art painting, Cartoon, and Sketch domains, respectively. For VLCS, V/L/C/S denote the VOC2007, LabelMe, Caltech, and SUN domains, respectively. For OfficeHome, A/C/P/R denote the Art, Clipart, Product, and Real World domains, respectively. For TerraIncognita, L100/L38/L43/L46 denote Location 100, Location 38, Location 43, and Location 46, respectively.

(a) PACS (A,C,P,S)

Method	A	C	P	S	Avg.
GroupDRO [24]	83.5	79.1	96.7	78.3	84.4
SAL† [15]	87.6	80.0	97.6	79.1	86.1
DR-DG † [18]	88.5	82.9	97.7	81.0	87.5
GLOT-DR† [21]	88.2	81.8	97.7	78.6	86.6
TRO† [23]	87.4	83.4	97.3	79.3	86.8
MEDRO † [12]	87.7	83.7	97.3	79.2	87.0
SR-WDRO† [16]	88.7	82.3	97.5	77.2	86.4
CS-DRO	**90.3**	**84.7**	**98.4**	**82.3**	**88.9**

(b) VLCS (C,L,S,V)

Method	C	L	S	V	Avg.
GroupDRO [24]	97.3	63.4	69.5	76.7	76.7
SAL† [15]	95.7	64.3	72.8	79.1	78.0
DR-DG † [18]	96.6	65.6	76.2	77.2	78.9
GLOT-DR† [21]	96.2	64.3	74.1	77.9	78.1
TRO† [23]	95.8	62.6	72.8	78.5	77.4
MEDRO † [12]	97.2	65.8	**76.6**	78.9	79.6
SR-WDRO† [16]	**97.9**	64.7	75.7	78.3	79.1
CS-DRO	97.6	**67.2**	76.5	**81.8**	**80.8**

(c) OfficeHome (A,C,P,R)

Method	A	C	P	R	Avg.
GroupDRO [24]	60.4	52.7	75.0	76.0	66.0
SAL† [15]	69.3	53.9	80.4	82.0	71.4
DR-DG † [18]	66.7	55.2	79.3	81.6	70.7
GLOT-DR† [21]	69.0	55.8	80.4	82.3	71.9
TRO† [23]	68.3	56.9	79.6	81.2	71.5
MEDRO † [12]	66.1	54.8	78.9	80.2	70.0
SR-WDRO† [16]	68.8	55.3	79.3	81.1	71.1
CS-DRO	**72.4**	**57.2**	**80.9**	**83.4**	**73.4**

(d) TerraIncognita (L100,L38,L43,L46)

Method	L100	L38	L43	L46	Avg.
GroupDRO [24]	41.2	38.6	56.7	36.4	43.2
SAL† [15]	59.9	47.0	59.5	42.4	52.2
DR-DG † [18]	54.0	46.5	59.3	41.7	50.4
GLOT-DR† [21]	56.2	45.4	55.8	43.2	50.2
TRO† [23]	57.2	46.9	59.8	41.2	51.3
MEDRO † [12]	55.8	43.3	59.0	38.5	49.2
SR-WDRO† [16]	56.0	**47.2**	58.8	40.7	50.7
CS-DRO	**62.8**	46.1	**60.4**	**47.1**	**54.1**

H.5 Computation cost

We compare the computational cost of CS-DRO with that of existing DRO-based methods. Methods in the WDRO family perform J-step iterative updates at each training step to approximately solve the inner maximization, which can significantly increase training time depending on the implementation. In particular, adversarial updates in the input space tend to incur high computation and GPU memory overhead; accordingly, recent approaches (including this work) prefer to perform DRO in the representation space. We measured for each method on PACS using a single NVIDIA RTX 3090: (i) the time per step, and (ii) the wall-clock time for the entire 5,000 training steps. The wall-clock time includes the overhead from evaluation, and we report the mean and standard deviation over three runs with different random seeds. The compared WDRO-based methods were reimplemented under the same training and evaluation protocol as our work for a fair measurement.

The results are shown in Table 9. Most methods have similar training times of around 0.20s per step, and total times on the order of 1000s for 5,000 steps. In contrast, SAL and GLOT-DR exhibit relatively higher time costs, which can be interpreted as resulting from additional OT-related iterative computations in addition to the inner maximization. CS-DRO shows a slight overhead (0.226s per step) compared to DR-DG or TRO, which is observed because additional computations such as proxy DAG estimation and structure-preserving constraints are included during training. However, compared to WDRO-based methods that use the same J-step inner solver, the overall computation cost of CS-DRO remains within a practical range.

I Full results of Table 1

Table 1 (main paper) in the main paper shows only the values averaged over the target domains (in each dataset) under the leave-one-domain-out evaluation. In contrast, Table 10 reports the accuracy (%) for each target domain under the same experimental setting, as a supplementary result to check whether the improvement in average performance is not biased to a specific domain but is achieved overall. From Table 10, we can see that CS-DRO achieves the highest performance on most target domains, or performance very close to the best. For example, in PACS it consistently achieves the top performance in all four target domains (A, C, P, S), and in VLCS and TerraIncognita it also maintains generally superior performance except in a few domains. This suggests that the improvement in average performance observed in the main text is not a phenomenon confined to a specific domain, but rather manifests broadly under various domain shifts.

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